

UNIVERSIDADE DE SÃO PAULO

Instituto de Ciências Matemáticas e de Computação

**Algorithmic Trading Enhanced by AI: Using News,
Geopolitical Events and NLP-Based Sentiment in Forex
Decision-Making**

Wellington Baltazar de Souza

Monograph - MBA in Artificial Intelligence and Big Data

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Wellington Baltazar de Souza

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*I dedicate this work to my family, for their unwavering support, encouragement, and belief in me
throughout this journey.*

*To my friends, who stood by me with patience, motivation, and understanding during both the
difficult and rewarding moments.*

To everyone who contributed, in any way, to the completion of this work, my heartfelt thanks.

“If I have seen further it is by standing on the shoulders of giants.”

Isaac Newton

ABSTRACT

BALTAZAR DE SOUZA, W. **Algorithmic Trading Enhanced by AI: Using News, Geopolitical Events and NLP-Based Sentiment in Forex Decision-Making**. 2026. 86 p. Monograph (MBA in Artificial Intelligence and Big Data) - Institute of Mathematics and Computer Sciences, University of São Paulo, São Carlos, 2026.

This work investigates whether NLP-derived sentiment and event features are capable of improving the risk-adjusted performance of a Forex trading strategy beyond a price-only baseline. The question matters because the Forex market is strongly driven by news flow and macroeconomic information, yet the literature on text-based prediction of financial returns has concentrated on equities, leaving the asset class comparatively underexplored. The proposed approach builds, on two major Forex pairs at minute resolution, a hybrid trading pipeline that combines technical signals with sentiment and event-based features extracted from financial text and from a public geopolitical-risk index, fused into a single trade decision under strict point-in-time discipline and a conservative publication-to-use delay; the precise sources, scorers and decision components are detailed in the methodology. Evaluation contrasts the hybrid against the price-only baseline out of sample under realistic transaction costs, complemented by an ablation that isolates the marginal contribution of the textual channel and by a formal statistical test of the difference in return distributions. The contribution is twofold: an empirical, falsifiable test of the hypothesis that adding NLP-derived textual features improves risk-adjusted performance on the two target pairs, and a reusable methodology that subsequent work can extend without reimplementing.

Keywords: algorithmic trading. forex. natural language processing. financial sentiment analysis. event extraction. FinBERT. GDELT. geopolitical risk. backtesting.

RESUMO

BALTAZAR DE SOUZA, W. **Algorithmic Trading Enhanced by AI: Using News, Geopolitical Events and NLP-Based Sentiment in Forex Decision-Making**. 2026. 86 p. Monograph (MBA in Artificial Intelligence and Big Data) - Institute of Mathematics and Computer Sciences, University of São Paulo, São Carlos, 2026.

Este trabalho investiga se atributos de sentimento e de eventos derivados de técnicas de Processamento de Linguagem Natural (PLN) são capazes de melhorar o desempenho ajustado a risco de uma estratégia de negociação algorítmica no mercado de câmbio (Forex) em relação a uma estratégia de referência baseada apenas em preços. A pergunta é relevante porque o mercado de câmbio é fortemente influenciado por fluxos de notícias e por informação macroeconômica; ainda assim, a literatura sobre previsão de retornos financeiros a partir de texto tem se concentrado em mercados acionários, deixando essa classe de ativos comparativamente menos explorada. A abordagem proposta constrói, sobre dois pares principais de câmbio em resolução de um minuto, uma estratégia híbrida que combina sinais técnicos com atributos de sentimento e de eventos extraídos de texto financeiro e de um índice público de risco geopolítico, integrados em uma única decisão de negociação sob disciplina temporal estrita (*point-in-time*) e com um atraso conservador entre a publicação e o uso da informação; as fontes de dados, os escoradores e os componentes de decisão são detalhados no capítulo de metodologia. A avaliação contrasta a estratégia híbrida com a estratégia de referência baseada apenas em preços, em dados fora da amostra, sob custos de transação realistas, complementada por um estudo de ablação que isola a contribuição marginal do canal textual e por um teste estatístico formal da diferença entre as distribuições de retornos. A contribuição é dupla: um teste empírico e falsificável da hipótese de que adicionar atributos textuais derivados de PLN melhora o desempenho ajustado a risco nos dois pares-alvo, e uma metodologia reutilizável que trabalhos subsequentes possam estender sem reimplementação.

Palavras-chave: negociação algorítmica. mercado de câmbio (Forex). processamento de linguagem natural. análise de sentimento financeiro. extração de eventos. FinBERT. GDELT. risco geopolítico. *backtesting*.

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LIST OF ABBREVIATIONS AND ACRONYMS

ADX	Average Directional Index (technical indicator)
ATR	Average True Range (volatility indicator)
BIS	Bank for International Settlements
BOJ	Bank of Japan
CAMEO	Conflict and Mediation Event Observations (event taxonomy)
ECB	European Central Bank
ECN	Electronic Communication Network
EUR	Euro (ISO 4217 currency code)
FOMC	Federal Open Market Committee
FX	Foreign Exchange
Forex	Foreign Exchange (market)
GBP	Pound Sterling (ISO 4217 currency code)
GPR	Geopolitical Risk (index of Caldara and Iacoviello)
HFT	High-Frequency Trading
JPY	Japanese Yen (ISO 4217 currency code)
MACD	Moving Average Convergence Divergence (technical indicator)
OHLC	Open, High, Low, Close (price-bar tuple)
OTC	Over-the-Counter (market)
P&L	Profit and Loss
pip	Percentage in Point (smallest standard FX price increment)
RSI	Relative Strength Index (momentum oscillator)
USD	United States Dollar (ISO 4217 currency code)
ZN	CBOT Ten-Year U.S. Treasury Note futures (ticker)
AI	Artificial Intelligence

BERT	Bidirectional Encoder Representations from Transformers
BERTimbau	BERT pre-trained for Brazilian Portuguese
BPE	Byte-Pair Encoding (sub-word tokenization algorithm)
CBOW	Continuous Bag-of-Words (word2vec architecture)
FinBERT	Financial-domain fine-tuned BERT
FinGPT	Open-source financial Large Language Model
GloVe	Global Vectors for Word Representation
GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit
LLM	Large Language Model
LSTM	Long Short-Term Memory (recurrent neural network cell)
ML	Machine Learning
MLM	Masked Language Modeling
NLP	Natural Language Processing
OOV	Out-Of-Vocabulary
OPRO	Optimization by PROMpting (LLM prompt-optimization method)
POS	Part-Of-Speech (tagging)
PLN	Processamento de Linguagem Natural (Portuguese for NLP)
RLHF	Reinforcement Learning from Human Feedback
RNN	Recurrent Neural Network
SHAP	SHapley Additive exPlanations (model-interpretation method)
SLM	Small Language Model
T5	Text-to-Text Transfer Transformer
TF	Term Frequency
TF-IDF	Term Frequency-Inverse Document Frequency
API	Application Programming Interface

CC-News	Common Crawl News (corpus)
CSV	Comma-Separated Values (file format)
FNSPID	Financial News and Stock Price Integration Dataset
GDELT	Global Database of Events, Language and Tone
GKG	Global Knowledge Graph (GDELT 2.0 table)
GPL	GNU General Public License
HTTP	HyperText Transfer Protocol
LEAN	open-source algorithmic-trading engine maintained by QuantConnect
LGPL	GNU Lesser General Public License
MVP	Minimum Viable Product
OANDA	online Forex and CFD brokerage (proper name)
RAM	Random-Access Memory
REST	Representational State Transfer
SQL	Structured Query Language
YAML	YAML Ain't Markup Language (data-serialization format)
ABNT	Brazilian Association of Technical Standards
ICMC	Institute of Mathematics and Computer Sciences (USP)
MBA	Master of Business Administration
NBR	Norma Brasileira (Brazilian Standard)
USP	University of São Paulo

LIST OF SYMBOLS

$P_{XY,t}$	Price of currency pair XY at time t
t	Discrete time index (minute bar)
H	Forecast horizon (number of bars into the future)
$\hat{p}_t$	Calibrated probability output of the meta-learner at time t
θ_{high}	Upper decision threshold on $\hat{p}_t$
θ_{low}	Lower decision threshold on $\hat{p}_t$
r_t	Trend-regime label at time t (bull, bear, sideways)
v_t	Event-veto flag from the news-context filter at time t
k	Number of folds in combinatorial purged k -fold cross-validation
X	Input token sequence to a Transformer block
Q	Query matrix in scaled dot-product attention
K	Key matrix in scaled dot-product attention
V	Value matrix in scaled dot-product attention
W^Q, W^K, W^V	Learned projection matrices for Q, K, V
d_k	Dimensionality of the key vectors (attention scaling factor)
h	Number of parallel attention heads in multi-head attention
n	Length of a sub-word n -gram in BPE tokenization

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1 INTRODUCTION

The foreign exchange market (Forex or FX) trades roughly USD 7.5 trillion a day across spot, swap and derivative instruments (Bank for International Settlements, 2022). A market of that magnitude does not respond to price charts alone. Currency pairs react, often within seconds, to central-bank decisions, military escalation, sanctions, supply-chain shocks and political instability. Market participants able to ingest, interpret and act on such textual information faster than the average competitor retain a measurable advantage.

Trading strategies that rely exclusively on technical indicators applied to OHLC price bars (Murphy, 1999) have been documented to perform adequately under stable price regimes and to deteriorate when news flow, rather than price flow, drives the market. Modern Natural Language Processing (NLP) and the public release of finance-tuned language models (Araci, 2019; Yang; Uy; Huang, 2020; Wu *et al.*, 2023; Yang; Liu; Wang, 2023) have substantially lowered the cost of revisiting that limitation empirically. The present work investigates a hybrid trading pipeline that complements price-based signals with sentiment- and event-based features extracted from text, evaluated under a strict backtesting protocol on historical data.

1.1 Theme and Problem

The theme is the integration of textual information (news articles, institutional reports, macroeconomic announcements) into algorithmic decision-making for Forex. The driving research question is the following:

Can NLP-derived textual features improve risk-adjusted Forex performance over a price-only baseline?

Hypothesis

The central premise of this study, that adding NLP-derived sentiment and event features to a competitive price-only baseline improves risk-adjusted Forex performance, is a *hypothesis*, not an established result. It is rendered plausible by the equity-market evidence reviewed in Section 1.2 and by the documented failure of a macro-only fusion on precisely the two target pairs (Helden, 2021); plausibility, however, is not proof. The empirical apparatus developed in this work exists to subject that premise to a falsifiable test rather than to presuppose it. Accordingly, the research question above is operationalized as a single falsifiable hypothesis:

H1 On EUR/USD and USD/JPY over the 2024 test span, the hybrid strategy of Chapter 3 produces a higher annualized Sharpe ratio than the price-only baseline trained on the same data through the same backtesting protocol, by a margin Δ that survives the empirical transaction-cost model of Section 3.8 and a statistical test of return-distribution difference

at the conventional significance level. Equivalently, the hybrid strategy delivers a smaller maximum drawdown under the same protocol. Either condition independently is sufficient to accept H1.

H1 is referenced under an explicit accept/reject criterion in Section 5.1 of the conclusion, derived from the evidence reported in Chapter 4. A rejection constitutes a valid and reportable outcome: it would establish that, under this protocol and on these pairs, the textual channel does not yield a statistically meaningful improvement over the price-only baseline. Such a negative finding is itself a contribution, and motivates stating the hypothesis in a form that the evidence can refute.

The problem decomposes into three sub-problems, each a distinct methodological challenge. First, a data problem: text and events at currency-grade scale must be collected from multiple sources, deduplicated, and time-aligned with FX prices (Leetaru; Schrodtt, 2013). Second, a representation problem: free-form text must be transformed into numeric features that retain financially meaningful signal (polarity, intensity, topical focus, uncertainty) without leaking future information into past decisions (Loughran; McDonald, 2011; Tetlock, 2007). Third, a decision problem: the combined feature space must drive entries, exits and position sizing in a manner that remains auditable and robust to regime change (López de Prado, 2018).

1.2 Justification and Motivation

Three lines of reasoning motivate this investigation.

The empirical case is decades old, although most of the supporting evidence comes from equities rather than Forex. Tetlock (2007), working on the US equity market, demonstrated that media tone has predictive content beyond what is contained in past prices. Caldara and Iacoviello (2022) constructed a geopolitical risk index from newspaper text and documented measurable real-activity and broad financial-market effects following GPR shocks. Both contributions — Tetlock on US equities and Caldara and Iacoviello on macroeconomic and broad-market outcomes — exemplify the dominant lines of text-based financial prediction; Forex itself has been studied predominantly through macroeconomic and microstructure variables, with comparatively little use of textual inputs. This asymmetry leaves room for an incremental contribution and motivates this work as a transposition of an equity-derived hypothesis into a Forex setting.

Two recent USP monographs sharpen this gap on both sides. On the model side, Felix (2024), produced in the same ICMC/USP MBA in Artificial Intelligence and Big Data as the present work, surveys seven machine-learning classifiers for swing-trading on the IBrX-100 and explicitly identifies textual information (“new financial articles, social media and even publications from economic analysts”) as a relevant qualitative input class that the implementation nevertheless leaves out. On the asset-class side, Helden (2021) builds a hybrid FX strategy that fuses a macro-fundamental signal (real interest-rate differential, GDP-growth differential and M2/reserves ratio) with a technical signal averaged from three moving-average crossovers, applied to twelve currency pairs. The hybrid produces an annualized Sharpe ratio above 1 on

all six emerging Latin-American pairs in the sample, but **drops to 0.24 on EUR and to -1.08 on JPY against the USD** (Helden, Tabela 6, p. 35). The two pairs on which the macro-only hybrid fails are precisely those targeted by the present work; the failure of the macro-only fusion on the developed-market pairs constitutes, in itself, a direct empirical motivation to replace the macro-fundamental channel with an NLP-derived textual channel.

The technological case is recent. Until 2018, financial text analytics relied predominantly on dictionary methods such as Loughran–McDonald (Loughran; McDonald, 2011), whose sensitivity is bounded by the contents of the lexicon. Transformer-based models (Vaswani *et al.*, 2017; Devlin *et al.*, 2019) and their finance-tuned descendants (Araci, 2019; Yang; Uy; Huang, 2020; Wu *et al.*, 2023; Yang; Liu; Wang, 2023) have advanced the achievable accuracy on financial sentiment well beyond the lexicon ceiling. Revisiting the Forex sentiment problem under this newer toolset is empirically tractable, and brings the resulting methodology closer to what is operationally feasible.

A third motivation is methodological reusability. Conducted as an applied dissertation within an MBA in Artificial Intelligence and Big Data, the present study is organized so that its methodology and pipeline remain extensible. Design decisions accordingly favor permissively-licensed components, an explicit interface boundary between perception, decision and execution, an auditable and configurable filter chain, and a perception layer decoupled from execution determinism so that model upgrades do not compromise reproducibility. The objective is a reusable methodology, not a one-off study: subsequent investigations should be able to refine the sentiment models, extend the asset universe or alter the evaluation protocol without reimplementing the apparatus.

1.3 Objectives

1.3.1 General objective

To design, implement and empirically evaluate a hybrid algorithmic-trading methodology for the Forex market, encompassing the data pipeline, the sentiment- and event-feature layers, the baseline and hybrid strategies and the backtesting protocol, applied to two major Forex pairs, EUR/USD and USD/JPY, and to determine, under controlled experimental conditions, whether the textual features yield a statistically meaningful improvement over a price-only baseline.

1.3.2 Specific objectives

1. **Construct a reproducible data layer.** Assemble at least a decade of intraday price history for the selected pairs, complemented by a multi-source historical corpus of news and event data (GDELT (Leetaru; Schrodt, 2013) as the primary source, with additional news archives integrated through a uniform adapter interface) and macroeconomic-announcement calendars, under strict point-in-time discipline.
2. **Define a sentiment- and event-feature layer.** Score textual and event information at

the same temporal granularity as price data, employing FinBERT-class language models (Araci, 2019; Yang; Uy; Huang, 2020) for sentence-level polarity, GDELT events for event-level features, and the Caldara–Iacoviello GPR index (Caldara; Iacoviello, 2022) as a continuous risk-regime signal.

3. **Specify and implement baseline and hybrid strategies in a shared framework.** The baseline draws on price-only feature families spanning technical indicators (such as moving averages, momentum oscillators and volatility measures), chart and candlestick patterns, and market-activity signals (such as volume and realized volatility). These are combined through per-family LightGBM sub-models whose calibrated probabilities feed a logistic meta-learner; the meta-learner output then drives a deterministic filter chain that emits BUY, SELL or HOLD on each bar. The hybrid retains the same architecture and adds a news- and event-feature family (FinBERT-class sentence-level scoring, GDELT events, and the Caldara–Iacoviello GPR index) as an additional sub-model in the meta-learner. Implementation proceeds in stages, from a minimum viable baseline up to the full architecture, with the exact family decomposition and filter set subject to revision based on experimental ablations (Section 3.7.1.1). Both strategies are evaluated within the same backtesting protocol, incorporating realistic transaction costs (spreads, slippage and rollover).
4. **Conduct a rigorous empirical evaluation.** Apply out-of-sample risk-adjusted metrics (Sharpe, Sortino, maximum drawdown, profit factor), formal statistical tests of return-distribution differences, and ablation analyses isolating the marginal contribution of the sentiment and event channels.

1.4 Document Structure

The remainder of this dissertation is organized as follows. Chapter 2 reviews the theoretical foundations: the structure of the Forex market, the principles of algorithmic and quantitative trading, the NLP concepts on which the methodology depends, the literature on geopolitical risk and event data, and recent advances in finance-tuned language models. Chapter 3 presents the methodology: data sources, feature engineering, model choices, backtesting protocol and evaluation criteria. Chapter 4 reports the experimental results, including the ablation analyses that isolate the contributions of the sentiment and event channels. Chapter 5 synthesizes the findings, discusses the limitations, and outlines directions for subsequent research.

At the present intermediate submission stage, the theoretical foundations and the methodology are presented in their definitive form; the experimental chapter will be completed for the final submission, once the empirical results, the metrics tables and the ablation analyses have been produced.

2 THEORETICAL FOUNDATION

This chapter reviews the conceptual building blocks this work relies on. The objective is not to provide an exhaustive survey of any of the underlying fields but to fix the terminology and the methodological assumptions that subsequent chapters use without further justification. The presentation is organized in five blocks: the structure of the foreign exchange market (Section 2.1), the principles of algorithmic and quantitative trading (Section 2.2), Natural Language Processing applied to financial text (Section 2.3), the use of geopolitical and event data in macro-financial research (Section 2.4), and recent hybrid designs that combine these pieces (Section 2.5).

2.1 The Foreign Exchange Market

The foreign exchange market is an over-the-counter (OTC), continuously-traded market in which currencies are quoted in pairs. The price of a pair XY at time t , denoted $P_{XY,t}$, is the amount of currency Y required to acquire one unit of currency X . Four characteristics differentiate Forex from equity markets and shape the methodology of the present work.

First, scale and liquidity. The 2022 BIS Triennial Survey reports an average daily turnover of USD 7.5 trillion across spot transactions, outright forwards, FX swaps, currency swaps and options (Bank for International Settlements, 2022), of which the spot segment alone exceeds USD 2.1 trillion. Such depth means that, for major pairs, even large algorithmic orders can typically be filled with negligible market impact, which is methodologically convenient: backtests in major pairs can ignore the price-impact term that distorts equity backtests at comparable notional sizes.

Second, decentralization. Forex has no central exchange. Liquidity sits with a handful of money-center banks and a growing number of electronic communication networks (ECNs). The consequence for empirical work is that “the price” at a given timestamp is not a single number but a distribution over venues; care is needed to reconcile feeds and to attribute realized slippage between strategy decisions and venue selection.

Third, leverage. Retail Forex brokers commonly offer leverage in the range 1:30 to 1:500, an order of magnitude higher than typical equity margin. Leverage transforms small price moves into large profit-and-loss swings; risk-management procedures and position-sizing rules therefore dominate strategy performance to a degree that is unusual in cash-equity research.

Fourth, the central role of macroeconomic information. Differential interest rates, balance-of-payments flows, central-bank communication, inflation prints and political stability are first-order drivers of cross-rate moves. In contrast with equity markets, where firm-level fundamentals and earnings revisions explain a substantial share of return variance, Forex is strongly exposed to macroeconomic and geopolitical news flows. The cross-asset evidence in Caldara and Iacoviello (2022) documents measurable real-activity and financial-market effects of geopolitical-risk shocks in general rather than Forex-specific effects, but the indirect implication for cross-rate moves is consistent with the structural characterization given here. This is precisely the structural

feature that makes Forex an interesting laboratory for an NLP-augmented strategy.

2.2 Algorithmic Trading and Quantitative Strategies

Algorithmic trading covers any approach in which buy and sell decisions are produced by a computer-encoded rule rather than by direct human discretion. Within that broad definition the literature commonly distinguishes between technical, fundamental and statistical approaches (Chan, 2008; Chan, 2013a).

Technical approaches treat past prices and volumes as the only information set and apply transformations (moving averages, oscillators, volatility filters, support-and-resistance heuristics) to derive entry and exit signals (Murphy, 1999). Their appeal is operational simplicity and low data requirements. Their weakness is that any signal extractable from past prices alone is in principle available to every other participant on the same data feed, and therefore prone to arbitrage decay.

Fundamental approaches use financial-statement data and macroeconomic series to estimate the equilibrium value of an asset and trade deviations between price and fundamental value. In Forex, fundamental analysis revolves around interest-rate parity, purchasing-power parity, current-account dynamics and central-bank policy reaction functions. The data is typically released at low frequency (monthly or quarterly), which makes fundamental approaches better suited to position trading than to intraday execution.

Statistical and machine-learning approaches replace closed-form rules with models fit to historical data. The financial-machine-learning literature (López de Prado, 2018; López de Prado, 2020) surveys the modern variants: ensembles of decision trees, deep neural networks, sequence models. LSTM networks and, more recently, transformer-based architectures (Vaswani *et al.*, 2017) are now standard tools for sequence modeling on financial time series.

A common methodological pitfall, emphasized by López de Prado (2018), is that financial data violates several of the assumptions on which standard cross-validation techniques rely: observations are not independent, distributions are non-stationary, and look-ahead leakage is easily introduced when features are computed without strict point-in-time discipline. Backtesting protocols, walk-forward evaluation and combinatorial purged cross-validation are therefore non-negotiable when claiming out-of-sample performance.

2.3 Natural Language Processing for Financial Text

This section reviews the NLP elements directly used by the methodology: how financial text becomes numeric features, how sentiment is scored on that text, how the Transformer architecture and its finance-adapted descendants supply the scorer, and how the orthogonal task of event extraction structures occurrence-typed signals from the same textual input. Generic-NLP background material (linguistic levels, static word embeddings, classical text-classification estimators, the full LLM training pipeline) is omitted: the focus is on the components this work

reuses.

2.3.1 Financial text representation

Financial text reaches a numerical representation through one of two families. The *bag-of-words* family represents a document by the multiset of its tokens (TF, TF-IDF, or presence/absence). The *distributional* family represents words by dense low-dimensional vectors learned from co-occurrence patterns under the *distributional hypothesis*: static variants assign one vector per word and conflate polysemy, while *contextual* embeddings produced by Transformer encoders such as BERT (Devlin *et al.*, 2019) yield a different vector per token occurrence. Polysemy resolution matters for financial text, where words such as *call*, *bond*, *liability* or *depreciation* carry technical senses distinct from everyday usage. The methodology of this work uses contextual embeddings from the FinBERT family of Section 2.3.3 for sentence-level scoring of financial text.

2.3.2 Sentiment analysis in finance

Sentiment analysis of financial text predates the deep-learning era. The lexicon-based approach of Loughran and McDonald (2011) introduces a domain-specific dictionary that corrects the misclassifications general-purpose lexicons commit on financial language: words such as *liability*, *tax* or *depreciation* carry negative polarity in everyday English but are neutral in a financial-statement context. Tetlock (2007) earlier showed that media tone, measured by dictionary methods of this kind, has predictive content for short-horizon equity returns beyond what is contained in past prices. Loughran–McDonald has remained an influential baseline; the FinBERT-class scorers reviewed below are the modern frontier and are benchmarked against it in this work.

2.3.3 Transformer models in finance

The Transformer architecture of Vaswani *et al.* (2017) replaced recurrent sequence models with *self-attention*, which computes for each token a weighted combination of every other token in the sequence; the context-sensitive representations and the training parallelism that this enables underlie nearly every modern language model. Two families derive from this architecture: *encoders* (the BERT lineage), suited to representation and classification, and *decoders* (the GPT lineage), suited to autoregressive generation. The methodology of this work uses encoders for sentence-level scoring.

For financial text, two finance-adapted encoders bear the *FinBERT* name. Araci (2019) fine-tunes a BERT base model on a small set of finance-specific sentiment datasets. Yang, Uy and Huang (2020) pre-trains BERT from scratch on a much larger corpus of financial communications and benchmarks the resulting model on the Financial PhraseBank dataset and other downstream finance tasks. Together they constitute the current methodological baseline for sentence-level sentiment scoring on English-language financial text.

Beyond the FinBERT baseline, the release of BloombergGPT by Wu *et al.* (2023) marks the entry of large foundation models into finance: a 50-billion-parameter decoder model trained on a curated mixture of public web text and Bloomberg’s proprietary corpus, outperforming general-purpose LLMs of similar size on financial-domain benchmarks. The open-source FinGPT initiative (Yang; Liu; Wang, 2023) addresses the reproducibility gap by providing open code, data-curation pipelines and fine-tuning recipes, and offers a practical path for academic and applied work that cannot rely on closed proprietary models.

More recent evidence narrows the gap between general financial NLP and Forex-specific sentiment modeling. Fatouros *et al.* (2023) evaluate zero-shot ChatGPT 3.5 on a curated dataset of Forex-related news headlines and report approximately 35 % better sentiment-classification performance and a 36 % stronger correlation between predicted sentiment and market returns than FinBERT on that dataset. The gap reflects model scale and pre-training data rather than an architectural limitation, a point reinforced by Ballinari and Maly (2025): domain-specific fine-tuning of contemporary open-source LLMs can match or exceed the zero-shot performance of general foundation models on FX sentiment tasks, and the resulting fine-tuned models outperform existing methods in both classification accuracy and the trading performance of sentiment-driven strategies. Outside FX but still within tradable financial sentiment, Iacovides, Zhou and Mandic (2025) introduce a finance-specific LLM aligned by Direct Preference Optimization and report state-of-the-art benchmark performance together with portfolio simulations that remain profitable under transaction costs. These results justify treating LLM-class scorers as a credible upgrade path beyond the FinBERT baseline rather than a speculative future extension; the cost–benefit trade-off against the inference-latency budget of any live deployment is discussed in the methodology chapter.

2.3.4 Event extraction from financial text

Where sentiment analysis aggregates the *tone* of a corpus, event extraction identifies the *occurrence* of specific events — a sanction package, a central-bank decision, a military operation, a sovereign downgrade — and attaches structured meta-data: actors, location, severity and time. The canonical NLP pipeline combines named-entity recognition (NER) for actor and location detection, event-trigger detection for the predicate, and role assignment for event arguments against a target ontology. For political and macroeconomic events the dominant ontology is **CAMEO** (Conflict and Mediation Event Observations), under which actors and event types are encoded into a fixed taxonomy that supports cross-document aggregation. The Global Database of Events, Language and Tone (GDELT), discussed in detail in Section 2.4, operationalizes this pipeline at scale: it processes broadcast, print and web news in more than a hundred languages and emits CAMEO-coded events updated every fifteen minutes. The separation of sentiment (tone) and event (occurrence) as orthogonal textual channels is a deliberate design choice of the methodology of this work.

2.4 Geopolitical Events and Macroeconomic News

While sentiment analysis aggregates the *tone* of a corpus, geopolitical and macroeconomic event analysis aims at identifying the *occurrence* of specific events (a sanction package, a central-bank decision, a military operation, a sovereign downgrade) and at attaching meta-data such as actors, location, severity and time. Two complementary data sources are referenced in this work.

The Global Database of Events, Language and Tone (GDELT), introduced by Leetaru and Schrodt (2013), monitors broadcast, print and web news in over a hundred languages and codifies events using the CAMEO taxonomy. The GDELT 2.0 release exposes three core tables, Events, Mentions and the Global Knowledge Graph (GKG), updated every fifteen minutes and freely downloadable. For the present work, GDELT serves as the primary high-frequency event-data source.

The Geopolitical Risk (GPR) index of Caldara and Iacoviello (2022) is a complementary source built from automated text analysis of the major newspapers' archives. The authors show that GPR shocks have measurable, negative and persistent effects on real activity and on financial variables. The combination of an event-level data source (GDELT) with a continuous risk index (GPR) lets the proposed framework distinguish between the impact of isolated events and the impact of sustained risk regimes.

Complementary cross-market evidence comes from Bampinas *et al.* (2026), who study stock and currency markets jointly and report that, under elevated geopolitical risk, stock–currency dependence strengthens and tail risks become more synchronized, especially in emerging markets. The result is not a trading rule and is not Forex-only; it is nonetheless directly relevant to the present work because it reinforces the view that geopolitical shocks alter the dependence structure around currencies rather than merely adding noise to a price-only process.

Recent evidence sharpens the empirical case for treating GPR as an investable signal in FX specifically. Liu and Zhang (2024) construct a zero-cost cross-sectional currency strategy that buys higher-GPR currencies and sells lower-GPR ones, sorting 42 currencies into five portfolios on monthly GPR indices over June 2002 to December 2019, and report a statistically significant excess return of 5.72 % per year before transaction costs and 4.73 % after bid–ask costs. The same study decomposes the GPR exposure into a country-specific idiosyncratic component and a regional component. A complementary result comes from Melone and Stathopoulos (2026), in a working paper that sorts currencies not on the *level* of geopolitical risk but on the *exposure* of their expected excess returns to the global geopolitical-threats index of Caldara and Iacoviello (2022): their dollar-neutral high-minus-low-exposure strategy yields an annualized return of 3.28 % with an annualized alpha of 2.76 % after controlling for the standard dollar and carry currency factors, and the corresponding traded factor is priced in the cross-section of currency returns. Notably, the same study finds that these geopolitical factor loadings are *unrelated* to the exposure of currency returns to measures of policy uncertainty, which supports the present work's decision to feed a dedicated geopolitical-risk index rather than substituting a generic economic-policy-uncertainty index for it. The two studies establish, by distinct constructions (a

level sort and an exposure sort), that GPR carries directional information for currency returns beyond its already documented effect on real activity and broad financial variables, which motivates including the GPR index as a continuous feature alongside the event-level features in the methodology of the present work.

2.5 Hybrid Approaches

Hybrid trading frameworks combine signals coming from heterogeneous sources (prices, volumes, order-book imbalances, sentiment, events) into a single decision rule. Two design questions distinguish proposals in the recent literature.

The first is the level of fusion. Early-fusion designs concatenate raw features into a single feature vector and let a downstream model learn the joint mapping. Late-fusion designs train a separate model per source and combine their outputs at a final stage, typically by stacking or by ensemble voting. Late-fusion is the safer default when the underlying sources have very different signal-to-noise ratios, as is the case between price-based features (low noise, high frequency) and text-derived features (high noise, lower effective frequency).

The second design question is the choice of language model. Until 2023 the methodological default for financial text was a fine-tuned BERT-class model such as the two FinBERT variants (Araci, 2019; Yang; Uy; Huang, 2020); with the release of BloombergGPT (Wu *et al.*, 2023) and the open FinGPT pipelines (Yang; Liu; Wang, 2023) the practical frontier has shifted toward larger foundation models, with measurable but uneven gains on financial benchmarks.

A third design question, raised by the most recent agentic-LLM proposals in finance, is the execution paradigm. The question is whether the language model produces structured features that feed a deterministic decision rule, or whether the language model is itself the decision-maker. According to the financial-NLP literature, the implicit default has been the first option: BERT-class scorers feed price-based models whose execution layer remains rule-based (Tetlock, 2007; Loughran; McDonald, 2011). The release of multi-agent LLM frameworks such as TradingAgents (Xiao *et al.*, 2024) explores the second option, in which ensembles of role-playing LLM agents (analyst, researcher, trader, risk manager) reach trading decisions through structured debate, and the agent ensemble itself emits the buy/sell signal. Subsequent proposals along the same lines specialize the agent ensemble for different objectives: ATLAS (Papadakis *et al.*, 2025) introduces dynamic prompt optimization through an Adaptive-OPRO mechanism around a central trading agent; TradingGroup (Tian; Salim; Xue, 2025) combines specialist agents with self-reflection and an automated data-synthesis pipeline for post-training; and QuantAgent (Xiong *et al.*, 2025) targets high-frequency trading by decomposing the price-driven technical-analysis layer into separate indicator, pattern, trend and risk agents.

The trade-off is clear. The agentic-execution variant is more expressive and easier to extend with new analysts. However, its outputs are non-deterministic, and the model drift introduced by vendor updates of the underlying foundation models makes rigorous backtesting under the validation discipline of López de Prado (2018) substantially harder. By contrast,

the deterministic-execution variant constrains the language model to a perception role, which preserves backtest reproducibility and regulatory auditability, at the cost of a less expressive decision-time pipeline.

The methodological intuition behind the present late-fusion design predates the modern NLP-era papers. Henderson (2002) (Ch. 10, Table 10.1, p. 204) sketches a manual “Signal Grid” in which a currency-pair decision is reached by combining four orthogonal channels (currency-economics indicator, capital-flow proxy, technical signal, and long-term valuation deviation) and converting the row of channel votes into a single Buy/Sell/Hold. That 2002 hand-built scheme is the conceptual ancestor of the deterministic execution chain of the present work; the present contribution is to replace the manual analyst with an algorithmic pipeline in which the “flow” column is filled by NLP-derived sentiment over GDELT and the “technical” column by a learned meta-learner over price features. Henderson also identifies the yen explicitly as a safe-haven funding currency that recovers when global risk appetite collapses (p. 56), an observation that motivates regime-aware feature engineering for USD/JPY distinct from EUR/USD in Chapter 3.

Empirical evidence for textual sentiment as an investable factor predates BERT-class scoring as well. Chan (2013b) (Ch. 6, p. 148) reports the Hafez and Xie (2012) portfolio sort on S&P 100 names using RavenPack sentiment scores: a long-top-decile / short-bottom-decile sentiment portfolio produced annualized returns of 52–156 % and Sharpe ratios of 3.9–5.3 over the 2008–2010 sample. The Sharpe magnitudes sit well above the range that the methodological discipline of López de Prado (2018) would treat as plausible without further scrutiny: the sentiment scorer of that period was lexicon-based, the portfolio was rebalanced daily without realistic costs, and the multiple-comparisons correction discussed in Section 3.10.1 of the methodology chapter was not applied. Nevertheless, the result is a useful empirical pedigree for the design choice of feeding a sentiment channel into a directional model. A replication on FX with FinBERT-class scoring, walk-forward validation and explicit transaction costs, which is what the present work undertakes for EUR/USD and USD/JPY, is expected to deliver a more modest but more defensible effect size.

FX-specific intraday evidence, while still sparse, points in the same direction. Hafez and Xie (2013) introduce a EUR/USD cross-over strategy driven by sentiment-inflection points and report that their U.S. macro sentiment index predicts intraday EUR/USD price moves for several hours after an inflection point. The present work does not reuse RavenPack’s proprietary indices or their exact cross-over rule; the paper is used here only as empirical support for preserving publication-time information at intraday resolution rather than collapsing all textual information to a daily frequency.

Hybrid designs targeting Forex specifically remain comparatively rare. The closest prior art at the time of writing is the recent work of Zhang (2025). The author builds daily sentiment indices from GDELT using a FinBERT scorer, feeds the indices into an XGBoost classifier and reports out-of-sample backtests on EUR/USD, USD/JPY and ten-year U.S. Treasury futures over the 2017–2025 window. The decision rule of that work is itself deterministic: the classifier output

is converted into a directional position by a fixed probability threshold and the resulting rule is evaluated unchanged on the held-out window. The methodological choices are close enough to those of the present work to warrant explicit positioning. The design overlap (sources, scorer, pairs) is high. However, the validation protocol (five-fold expanding-window cross-validation), the temporal resolution (daily), the absence of a chart-and-candlestick pattern layer and the thin nature of the deterministic execution stage (a single-threshold rule) are points of divergence. The implications are discussed in Section 3.1 of the methodology chapter.

The cost-adjusted Sharpe ratios reported by Zhang (2025) are 5.87 on EUR/USD and 4.65 on USD/JPY. Obviously, such figures sit well above the range that the methodological discipline of López de Prado (2018) would treat as plausible without further scrutiny. In this regard, the comparison with that benchmark is one of the reporting commitments of Chapter 4.

For the present work the implications are threefold. First, the methodology adopts a late-fusion architecture, with FinBERT-class models for sentence-level sentiment and a separate event-extraction layer fed by GDELT and the GPR index. Second, the execution paradigm is deterministic: language models occupy the perception layer only, and trading decisions are emitted by a rule-based stage on top of cached perception outputs. Third, where performance gains and computational budget justify it, larger foundation models are evaluated as drop-in replacements for the sentence-level scoring component, with explicit measurement of the inference latency cost. The latter is a constraint that purely academic studies typically ignore but that any future live deployment requires the strategy to honor.

Table 1 positions the present work against the closest recent proposals in the AI-augmented trading strand.

The present work is positioned closest to Zhang (2025) in source coverage and asset class, and furthest in validation discipline and execution layer. The methodological divergence is elaborated in Section 3.1 of the methodology chapter.

Table 1 – Recent hybrid trading proposals positioned against the present work.

Work	Domain	Perception / Validation	Execution	
		scorer		
Zhang (2025)	Forex (EUR/USD, USD/JPY) + UST	FB → XGBoost on daily GDELT	5-fold expanding	Single threshold rule
Xiao <i>et al.</i> (2024)	US equities	Multi-role LLM analyst debate	WF	Agentic decision
Papadakis <i>et al.</i> (2025)	US equities	LLM + Adaptive-OPRO prompt-opt.	WF	Agentic decision
Tian, Salim and Xue (2025)	US equities	Specialist agents + self-reflection	WF	Agentic decision
Xiong <i>et al.</i> (2025)	HFT	Decomposed indicator agents	Tick-level	Agentic decision
This work	Forex (EUR/USD, USD/JPY)	FB + LM dictionary + GPR + GDELT events	CPK + embargo + WF	Late fusion + 7-filter deterministic chain

Acronyms: CPK = combinatorial purged k -fold; WF = walk-forward; FB = FinBERT; LM = Loughran–McDonald.

3 METHODOLOGY

This chapter sets out the methodology adopted by the present work, translating the conceptual framework reviewed in Chapter 2 into a concrete experimental pipeline. The presentation is organized in eleven sections: the positioning of the present work relative to the closest prior art (Section 3.1), an overview of the research design (Section 3.2), the data architecture and storage layer (Section 3.3), the Forex price-data pipeline (Section 3.4), the news and event-data pipeline (Section 3.5), the sentiment and event feature layer (Section 3.6), the fusion architecture (Section 3.7), the backtesting and evaluation protocol (Section 3.8), the execution engine (Section 3.9), the reproducibility procedures and threats to validity (Section 3.10), and a closing summary (Section 3.11).

The methodology is deliberately empirical and engineering-led. Every design choice in this chapter is motivated either by a constraint exposed in Chapter 2 (for example, the methodological discipline imposed by financial machine learning (López de Prado, 2018)) or by an operational consideration (storage budget, license compatibility, reproducibility). Choices that follow from neither are recorded as open items rather than asserted.

3.1 Positioning Relative to Prior Art

The closest published work to the present methodology is the recent study of Zhang (2025), introduced in Section 2.5. The design overlap with the present work is substantial. Both works share the same primary text source (GDELT), the same sentence-level scorer family (FinBERT), the same two major Forex pairs (EUR/USD and USD/JPY) and a near-identical evaluation window (2017 to 2025 in that study, 2015 to 2024 in the present one). For this reason, the methodology cannot be presented as a self-contained engineering exercise without first declaring how it differs from that benchmark. Six points of divergence are stated up front. Each one is motivated by a constraint that subsequent sections of this chapter elaborate.

Validation protocol. Zhang (2025) adopts a five-fold expanding-window cross-validation scheme on daily features and reports cost-adjusted Sharpe ratios of 5.87 on EUR/USD and 4.65 on USD/JPY. Sharpe ratios in that range, sustained across multiple assets on out-of-sample data, sit above the upper bound that the methodological discipline of López de Prado (2018) treats as plausible without explicit correction for selection bias and backtest overfitting. According to that discipline, the present work adopts the more conservative combinatorial purged k -fold cross-validation in Section 3.8, with an explicit embargo period between training and validation folds and a single held-out test pass per strategy variant. The trade-off is accepted in advance: lower reported Sharpe figures than Zhang (2025), in exchange for stronger out-of-sample credibility.

Temporal resolution. The scorer in Zhang (2025) aggregates GDELT articles into daily indices and the trading model emits daily decisions. The present work runs the price layer at minute resolution. Sentiment features are produced per article and bucketed onto the minute grid

using each article’s publication timestamp (Section 3.6.4), so the intraday arrival pattern of news is preserved on the sentiment side. Event-aggregate features (CAMEO category counts, GPR regime) remain at daily resolution and are forward-filled onto the minute grid (Section 3.6.5); the gain over Zhang (2025) on those features is in the price-execution wrapper around them, not in their own arrival timing. The asymmetry is documented as a feature-engineering choice rather than asserted as a uniform intraday upgrade.

Feature breadth on the price side. Zhang (2025) does not include chart-and-candlestick pattern features or a structured technical-analysis stack on top of the price input. The present work adds the four-family price input space described in Section 3.6: technical-analysis structural features, indicator features, candlestick-pattern detectors from (Nison, 1991) via the TA-Lib reference implementation, and classical chart patterns following the kernel-regression detection procedure of Lo, Mamaysky and Wang (2000). The motivation is the position taken in Section 2.5: hybrid strategies should evaluate the marginal contribution of textual features against a price-only baseline that is itself competitive, rather than a degenerate one.

Feature breadth on the text side. In addition to the FinBERT scorer, the present work also runs a Loughran–McDonald dictionary scorer (Loughran; McDonald, 2011) as a transparent baseline. Furthermore, the GPR continuous regime index of Caldara and Iacoviello (2022) complements GDELT’s discrete event records, and additional corpora (FNSPID (Dong; Fan; Peng, 2024), CC-News, central-bank releases and the Reddit Pushshift archive (Baumgartner *et al.*, 2020)) are ingested through the adapter pattern described in Section 3.5. The dictionary baseline is retained as a sanity check on the transformer scorer in the spirit of the methodological caution recommended by López de Prado (2018). The additional corpora reduce the single-source dependence that a GDELT-only design carries.

Decision architecture. The execution layer in Zhang (2025) is itself deterministic but thin: the XGBoost probability is converted into a directional position by a fixed threshold, and that single-rule layer is evaluated unchanged out of sample. The present work keeps the deterministic-execution commitment but enriches the layer in two directions (Sections 3.7.1 and 3.7.2). First, a small meta-learner combines the calibrated outputs of four sub-models into a single probability rather than relying on a single classifier. Second, a configurable filter chain converts that probability into a trade decision through a sequence of named filters (trend-regime, indicator, pattern, news-context, risk-guard, capital-management, terminal threshold), and each trade carries an audit trail consisting of the contribution of every filter. The architectural commitment is documented in Section 3.7 as “agentic perception, deterministic execution”. The divergence is therefore a thicker, multi-filter deterministic execution layer rather than a presence-versus-absence contrast with Zhang (2025).

Execution path and license. Zhang (2025) stops at the backtest stage and reports no live-execution path. The present work runs its backtests on the open-source LEAN engine of QuantConnect, under the Apache 2.0 license, and reuses LEAN’s official OANDA brokerage adapter as the eventual live-trading path (Section 3.9). As recorded in Section 1.2 of the in-

roduction, the motivation is the author’s intention to return to live algorithmic trading after the MBA. In this regard, a permissive license and an official brokerage adapter make the path from backtest to paper trading to live trading an exercise in re-targeting the runtime rather than re-implementing the strategy.

The remaining sections of this chapter elaborate the choices summarized above. The empirical comparison against Zhang (2025) on the metrics in common (Sharpe ratio, maximum drawdown, hit rate, turnover) is reported in Chapter 4.

3.2 Research Design

The research design is operational rather than theoretical: it specifies what is to be built, on which data, and how the resulting artifact is to be evaluated. The general objective stated in Section 1.3, "to develop and evaluate a hybrid algorithmic trading strategy that combines technical, sentiment-based and event-based signals on Forex pairs", decomposes into five concrete deliverables. The empirical evaluation is conducted exclusively on the foreign-exchange market, which is the deliberate scope of this work; the architecture, however, is designed to be asset-class agnostic, so that the market-specific constructs are isolated behind a common instrument abstraction and a set of configurable filters. Adapting the same pipeline to other financial markets (equities, ETFs, bonds, futures, commodities or cryptocurrencies) is therefore, in principle, an exercise in adapter substitution and configuration rather than redesign; this is developed as a direction for future work in Section 5.2.9.

- **Build a reproducible data pipeline.** Bulk-ingest historical Forex price data and a heterogeneous set of textual corpora into a single canonical storage layer. The pipeline must be re-runnable end-to-end without manual intervention.
- **Construct a feature layer.** Derive technical indicators from price data, sentence-level sentiment scores from financial text, and event-level scores from geopolitical-event corpora. Align all features to a common minute-level grid.
- **Train and backtest a baseline strategy.** Implement a price-only strategy as a control. Evaluate it under a strict walk-forward, purged cross-validation protocol (López de Prado, 2018).
- **Train and backtest a hybrid strategy.** Add the sentiment and event features to the baseline, using a late-fusion architecture (Section 2.5), and evaluate it under the same protocol.
- **Report the comparison honestly.** Report Sharpe ratio, maximum drawdown, hit rate and turnover for both strategies, in-sample and out-of-sample, with explicit discussion of the threats to validity.

Figure 1 summarizes the five deliverables as a single data-flow diagram. Heterogeneous data sources (left column) are ingested by self-registering `NewsDataSource` adapters, normalized into a single Parquet store partitioned by year and month, and converted into the five-family feature space (technical-analysis structural, indicator, pattern, news, market-activity) consumed by five LightGBM sub-models. A logistic meta-learner combines their calibrated outputs into $\hat{p}_t$, and a deterministic chain of filters then converts $\hat{p}_t$ into the trade decision, with every filter contribution persisted in an audit trail. The diagram embodies the methodological commitment of Section 3.7, *agentic perception, deterministic execution*: the perception layer (blue dashed box) is allowed to incorporate non-deterministic models because its outputs are precomputed once and read deterministically at backtest time, whereas the execution layer (red dashed box) is rule-based and reproducible regardless of upstream non-determinism. Backtests run on the LEAN engine under the combinatorial purged k -fold protocol of López de Prado (2018), and report Sharpe, maximum drawdown, hit rate, average holding time and turnover.

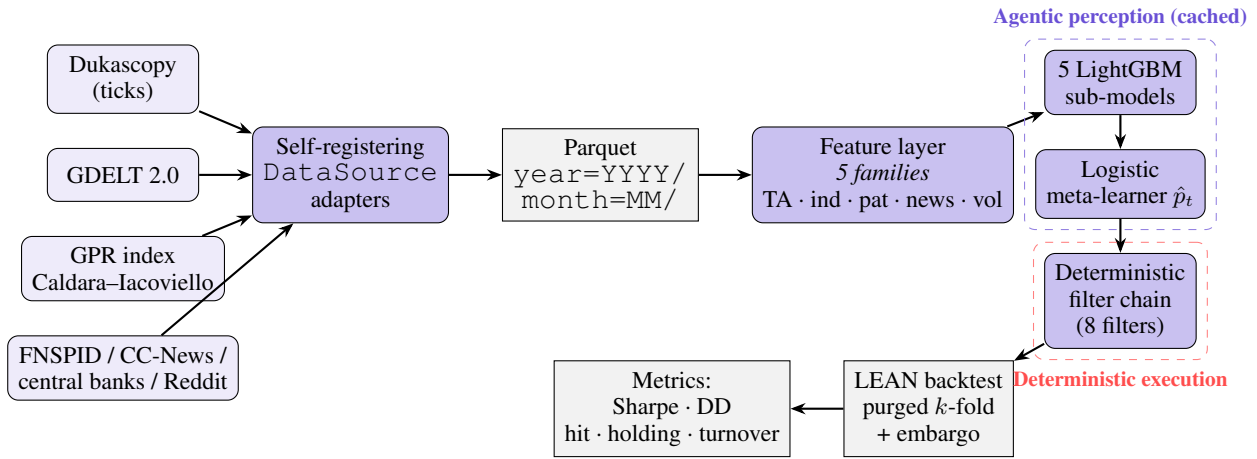


Figure 1 – System pipeline.

3.2.1 Experimental Variables and Type of Relationship

For clarity and to align the description with the methodological vocabulary of the MBA Methodology course, the experimental variables are stated explicitly. Table 2 groups them by role.

Type of relationship investigated. The present work conducts a *controlled comparison* (ablation) between two strategy variants on common data and a common protocol. It does not assert causal mechanisms by which textual signals affect FX rates; it measures whether the marginal predictive content of those signals translates into measurable risk-adjusted return improvement after transaction costs.

The currency pairs studied are EUR/USD and USD/JPY. Two pairs are deliberately included rather than one. The Japanese yen quotes the pip at the second decimal place rather

Table 2 – Experimental variables of the present work.

Role	Variables
Independent (manipulated)	Feature families of Section 3.6 (TA-structural, indicators, candlestick and chart patterns, news); LightGBM hyperparameters restricted to the small explicit grid of Section 3.10; calibrated thresholds θ_{high} and θ_{low} of Section 3.7.2; presence or absence of the news sub-model : the key manipulation isolating baseline from hybrid.
Dependent (measured)	Annualized Sharpe ratio (after transaction costs); maximum draw-down; hit rate; average holding time; turnover. All as defined in Section 3.8.
Controlled (fixed)	Empirical cost model (same spread and slippage distributions for baseline and hybrid); temporal split (2015–2022 / 2023 / 2024); random seeds documented in Section 3.10; pinned versions of NumPy, LightGBM, scikit-learn and PyArrow declared in <code>requirements.txt</code> .

than the fourth and exhibits a distinct volatility regime as a carry-trade vehicle, so building for both pairs forces a clean abstraction in the price downloader, the feature layer and the backtester. A single-pair codebase tends to bake quote-convention assumptions into hard-coded constants.

3.3 Data Architecture and Storage

Terminology. This work deliberately adopts the vocabulary of the LEAN execution engine (Section 3.9) as its system-level standard, precisely because LEAN is the platform on which the empirical work runs: aligning the written methodology with the engine’s own concepts removes the translation gap between the dissertation and the running system, and, because LEAN is itself multi-asset, keeps the present Forex study expressible in the same terms as the asset-class-agnostic architecture of Section 3.2. Under this convention a *symbol* identifies a tradable instrument, a *security type* names its asset class, a *QuoteBar* is the bid/ask bar at a given *resolution* (tick, minute, daily), and a *market* names the venue. The Forex parallels are stated where helpful: here a symbol is a currency pair (for example EUR/USD), the security type is *forex*, and the price-movement unit is the pip.

The whole experimental pipeline rests on a data-architecture decision: *Apache Parquet is the canonical source of truth for every dataset, regardless of source, while execution-facing derived data is materialized once into a durable engine-native store and reused.* The distinction is deliberate. Some datasets are read a handful of times for analysis, joining and model training; others sit on the hot path of the backtester, which replays the same historical inputs across the many simulations of a parameter and ablation sweep. Forcing both classes into a single format would be a mistake: the cost of the experimental work is not downloading or normalizing once, but executing thousands of backtests, so the dominant optimization target is replay efficiency. The architecture therefore splits storage by reuse rather than imposing one format everywhere.

Three properties motivate Parquet as the canonical layer.

First, engine portability. Parquet is a columnar format with mature readers in every language relevant to this work (Python, C#, Rust, Java). Any future replacement of the backtester, of the model-training stack or of the analysis layer can read the same files without conversion. A backtester-specific format, by contrast, would couple the data architecture to one engine.

Second, queryability. The DuckDB analytical engine reads Parquet directly and executes SQL over the partitioned tree without any ingestion step. Joining sentiment scores to price bars at minute resolution, computing coverage statistics for the news corpora, or producing exploratory plots reduces to a single query rather than a custom loader. The same Parquet tree therefore serves both the backtester and the analytical workflow.

Third, reproducibility. The Parquet tree is the source of truth; every other on-disk or in-memory artifact is derived. If a derived artifact is lost, it is rebuilt from Parquet in seconds. If the Parquet tree itself is lost, it is rebuilt from raw payloads via the ingestion pipeline. There is no ambiguity about which version of which dataset a given backtest used.

The on-disk layout follows a partitioned convention. Forex price data lives at `parquet/forex/<pair>/year=YYYY/month=MM/data.parquet`, where `<pair>` is one of `eurusd` or `usdjpy`. News and event data follow the analogous `parquet/<source>/<table>/year=YYYY/month=MM/data.parquet` layout already established by the news-ingestion sub-project. Derived sentiment scores live at `parquet/sentiment/` and event scores at `parquet/events/`, each partitioned in the same way.

The backtester, discussed in Section 3.9, expects its own resolution-bucketed file format, which it replays on every simulation. Rather than reconvert from Parquet on each run, a recurring cost in CPU and input/output that would be paid thousands of times across a sweep, the present work materializes an engine-native execution store once, at `lean-data/`, and reuses it across all runs. This store is durable: it is written to the filesystem, survives between sessions, and is rebuilt only when it is missing or its version is stale. It remains a *derived* artifact (reproducible from the canonical Parquet tree, so Parquet stays the single authoritative copy) but it is not regenerated per run, which is the distinction that matters for replay efficiency. For workstations with spare memory, a copy may optionally be staged on a Linux `tmpfs` mount at `/dev/shm/lean-data/` to serve reads from main memory; that copy is an accelerator re-stageable from the durable store, not the source of truth. Only minute-resolution price and the consolidated decision features are materialized this way. Tick-level data, which would otherwise contribute the bulk of the storage footprint, is kept only in Parquet (approximately five to ten gigabytes for two pairs over ten years) and is consumed during slippage modeling but not during normal backtest runs.

A second decision concerns *when* derived features are computed, and it is what keeps the backtest consistent with eventual live execution. Feature and sentiment computation is a single, mode-agnostic engine (a function of the current bar and the rolling state it carries) run vectorized over history during backtesting and one bar at a time in live trading. A single implementation is non-negotiable: two would expose the deployed model to inputs it was never trained on, the

classic train-and-serve skew. Its outputs pass through a read-through cache: a request returns the cached value on a hit and otherwise computes it, stores it, and returns it. The consequence is that every bar is computed *exactly once*, the first time it exists, and every later access is a cache hit. During a backtest sweep the first run fills the cache and the remaining runs (which vary only the filters, thresholds and ablations over the same history) read from it, eliminating the repeated recomputation that dominated the naive design. In live execution the same engine and the same cache are used: warm-up history and post-restart state are cache hits, and the only genuine computation is the bar that has just closed, because it did not exist before. The model itself is always trained offline on the historical panel and loaded for inference; production never trains. The cache backend is left open behind a port abstraction: an in-process least-recently-used structure and a file-backed store suffice for the present work, with a higher-performance backend available as a later, optional upgrade without any change to the calling code.

3.4 Forex Price-Data Pipeline

Historical price data for EUR/USD and USD/JPY is to be obtained from Dukascopy¹, a Swiss broker that publishes free tick-level historical data for major Forex pairs back to 2003 through a public HTTP datafeed. Retrieval is a direct HTTP download from that datafeed: each hour of tick data is a compressed binary file fetched over HTTP, requiring no account, no credentials and no third-party client library. The choice of Dukascopy over the available alternatives reflects the constraints of the present work: the data must cover the full target window (2015-02 to 2024-12), it must include both pairs, it must be free for academic use, and bulk download must be possible without rate limits prohibitive to a ten-year retrieval. HistData covers the window but only at minute resolution; the OANDA v20 application programming interface offers approximately five years of rolling history; Interactive Brokers is paid and rate-limited; TrueFX has a smaller pair set. Dukascopy is the only option that satisfies all four constraints. As a resilience measure, a lightweight daily-resolution fallback adapter (Yahoo Finance via the `yfinance` package) is retained so that, should the Dukascopy datafeed be unavailable for a given month, the pipeline continues at reduced fidelity rather than stalling; the fallback is not the primary backtest input.

The choice to commit to a single venue, rather than a synthetic best-bid/best-ask composite across multiple ECNs, follows Halls-Moore (2015) (Ch. 3, §3.3.3, p. 19): retail Forex brokers are not obligated to share trade prices with each other or with central aggregators, so any composite feed reconstructed from public sources differs from the executable price available at a given broker. The present work therefore commits to the Dukascopy bid/ask stream throughout the pipeline; the transaction-cost model of Section 3.8 is calibrated against the same venue's realized-spread distribution, so that backtest costs and live-execution costs are sourced from a consistent quote stream.

¹ <https://www.dukascopy.com/>

A second constraint specific to FX is that the total return of a multi-day position includes the rollover swap differential, not only the directional price move. Chan (2013b) (Ch. 5, eq. 5.6, pp. 113–114) makes this explicit: ignoring rollover on overnight FX positions is equivalent to a look-ahead error of opposite sign to the rate differential, and the magnitude can dominate the daily directional return on low-volatility pairs. The present work therefore models rollover as part of the daily P&L: for any position held across the New York 17:00 ET roll, the carry contribution is computed from the broker-published swap quote for the pair on that date, with the standard convention that the pair-side rate differential determines the sign. The convention is recorded in the unit tests of the backtester and is identical for the baseline and for the hybrid strategy, so that the comparison reports the marginal contribution of textual features net of identical carry treatment.

The price-ingestion code is planned as a sibling project to the existing news-ingestion code, reusing the same self-registering adapter pattern. A single market-data abstraction defines four operations (`plan`, `is_done`, `execute`, `unit_key`) and each concrete source (Dukascopy first, with HistData and TrueFX available as cross-checks) is to register an implementation under a stable name. The unit of work is one (pair, year, month) tuple, as in the news pipeline; idempotency is guaranteed by inspecting Parquet file existence on disk. The implementation status of the price-ingestion sibling at the time of submission is reported in Chapter 4.

Each hour of ticks is downloaded over HTTP as a compressed binary blob from the Dukascopy data servers and written first, unmodified, as a raw payload under `raw/dukascopy/...`; decoding is deferred to the transform stage, which keeps fetching and normalizing as separate, separately re-runnable steps. The transform stage then decompresses and decodes the blobs into bid/ask quote records, writes them as Parquet under the canonical layout, and resamples them into minute-resolution `QuoteBar` records (open, high, low, close on both bid and ask, plus tick count) at `parquet/forex/<pair>/...` alongside the tick partition. The minute-resolution Parquet is the input to all downstream analysis and feature engineering, and the source from which the durable `lean-data/` execution store is materialized once for replay (Section 3.3). Figure 2 shows this price path end to end.

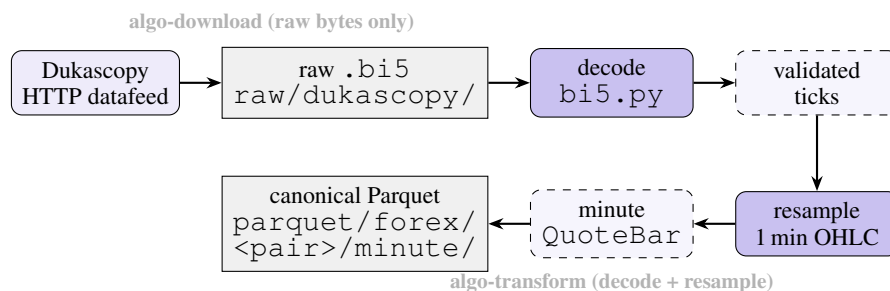


Figure 2 – Forex price data pipeline: raw Dukascopy bytes (`algo-download`) decoded and resampled into canonical minute `QuoteBar` Parquet (`algo-transform`).

i.i.d. assumption

Minute-resolution Forex returns are **not** independent and identically distributed. They exhibit serial autocorrelation in absolute returns (volatility clustering), regime-dependent unconditional means during macroeconomic-announcement windows, and a fat-tailed conditional distribution that varies across the session. Naive k -fold cross-validation, which presumes exchangeability of training and test samples, leaks future information into the past and inflates in-sample metrics. This work mitigates the violation in two ways: by adopting the combinatorial purged k -fold scheme with an embargo period (Section 3.8), which removes feature observations whose forward label horizon overlaps the validation fold, and by walk-forward training, which preserves the chronological order of the data at every refit. Neither mitigation makes the data i.i.d.; together they avoid the specific failure modes that the i.i.d. assumption would create.

3.5 News and Event-Data Pipeline

The news side of the pipeline ingests several corpora, each as an independent self-registering adapter. The order in which adapters are built reflects an explicit relevance ranking for Forex.

3.5.1 Source priority and rationale

The Forex literature reviewed in Sections 2.1 and 2.4 attributes the largest cross-rate moves to monetary-policy communication and to broad geopolitical shocks. The corpora are therefore prioritized by their proximity to these two drivers.

Central-bank releases (FOMC, ECB and BOJ) come first. The volume is small but the signal-to-noise ratio is the highest of any corpus available: a single sentence in an FOMC statement routinely moves USD/JPY by tens of pips. The planned adapters scrape the official archives directly and store one document per release. Implementation status at the time of submission is reported in Chapter 4.

GDELT 2.0 (Leetaru; Schrod, 2013) is implemented in the news-ingestion sub-project at the start of the methodology phase, in the form documented in Section 3.3. It provides the broadest coverage of the target window through three tables (Events, Mentions, and the Global Knowledge Graph) updated every fifteen minutes, and is consumed unchanged.

The **Geopolitical Risk index** of Caldara and Iacoviello (2022) is downloaded as a single CSV from the authors' public site². It is a derived index rather than a raw corpus, but its inclusion is methodologically important: GPR provides a continuous regime measure that complements the discrete event records of GDELT.

The **FNSPID** corpus of Dong, Fan and Peng (2024) is a recent large-scale financial-news dataset combining 29.7 million stock-price records with 15.7 million time-aligned news articles spanning S&P 500 companies from 1999 to 2023. Its bias is towards US equities, but

² <https://www.matteoiacoviello.com/gpr.htm>

the macroeconomic coverage it contains (Federal Reserve commentary, oil and dollar shocks, geopolitical reactions) carries information for USD-denominated pairs. It is ingested via the Hugging Face datasets interface as a one-shot bulk pull.

The **Common Crawl News** dataset (CC-News)³ provides general-news coverage from August 2016 onwards. It adds breadth at the cost of a higher noise floor, since most articles in any given month are unrelated to Forex; relevance filtering is therefore part of the feature-extraction step rather than the ingestion step.

The **Reddit archive** is consumed via the Pushshift dumps documented by Baumgartner *et al.* (2020). Three subreddits are retained for relevance (`r/Forex`, `r/economics` and `r/wallstreetbets`) covering the period 2008 to 2023.

Two further sources are listed as nice-to-have additions, scoped for inclusion only if the build budget allows. The **Trump Twitter Archive**⁴ provides every public tweet from the accounts of Donald J. Trump between 2009 and 2021 in structured form. It serves as a single-influencer corpus for the 2017–2021 period, during which Trump’s tweets demonstrably moved USD/CNY, USD/JPY and EUR/USD on multiple occasions through tariff threats, criticism of the Federal Reserve and trade-negotiation announcements. **StockTwits**⁵ provides retail-trader chatter with a finance-domain bias and is accessed via its public stream.

3.5.2 Excluded sources

Live access to Twitter/X is excluded. Following the platform’s policy changes from 2023 onwards, the API tiers compatible with bulk historical retrieval moved out of the budget range realistic for an academic project, and unauthenticated scraping is both technically blocked and prohibited by the platform’s terms of service. The present work treats this as a documented scoping decision rather than a methodological gap; the social-media tier is supplied by the Reddit archive, by StockTwits and, for the 2017–2021 sub-period, by the Trump Twitter Archive.

3.5.3 Coverage rule for the training window

The target time window is 2015-02-19 to 2024-12-31, ten years, with the lower bound fixed by the GDELT 2.0 launch date. Within this window the actual training span is determined empirically by a coverage rule: the training window is the largest contiguous span over which at least two news corpora are active for at least eighty per cent of the constituent months. The rule is computed from the ingested data after the first phase of the pipeline finishes, and the resulting coverage matrix (corpora $\times$ month) is reported as a figure in the experimental chapter. The rule replaces an arbitrary ex-ante cut-off with a reproducible criterion that survives the eventual addition or removal of a source.

³ Common Crawl Foundation. <https://commoncrawl.org/blog/news-dataset-available>.

⁴ <https://www.thetrumparchive.com/>

⁵ <https://stocktwits.com/>

3.5.4 Dataset Summary

Table 3 aggregates the sources of Sections 3.4 and 3.5 into a single descriptive view. Row counts are order-of-magnitude estimates derived from the public documentation of each source; the exact counts after the coverage rule of Section 3.5.3 is applied are reported in Chapter 4.

Table 3 – Sources used in the present work, with target window and known biases.

Source	Window	Approx. rows	Cadence	Known bias / limitation
Dukascopy ticks (EUR/USD, USD/JPY)	2015–2024	~5–10 GB / pair	Tick	Quote-only; no order-book depth
GDELT 2.0 Events / GKG	2015–2024	~1.0 B rows	15 min	English-source over-representation; publication-time lag
GPR index (Caldara–Iacoviello)	2015–2024	~2.5 k daily rows	Daily	US-newspaper basket; geopolitical only
FNSPID	2015–2023	15.7 M articles	Daily	US-equities bias; macro coverage indirect
CC-News	2016–2024	~50 M articles	Hourly	General news; high noise floor
Central-bank releases (FOMC, ECB, BOJ)	2015–2024	~10 k items	Per-meeting	Low volume, very high SNR
Reddit (r/Forex, r/economics, r/wallstreetbets)	2015–2023	~10 M posts	Continuous	Subreddit selection bias; retail-trader skew

Class balance for the directional label (BUY / HOLD / SELL) and the fraction of missing data per source are reported in Section 4.2 of Chapter 4 once the ingestion has settled. The bias column above is the qualitative counterpart of the quantitative coverage matrix reported there.

3.6 Feature Layer

The feature layer maps raw price and text data into the input matrix consumed by the strategy model. The decomposition adopted here follows the five-family input space used by a discretionary Forex trader: technical-analysis features describing market structure, indicator features quantifying momentum and volatility, chart and candlestick pattern features triggering entries, news features supplying context and risk vetoes, and market-activity features capturing the relative trading activity of each currency. The first three families and the market-activity family are derived from the price Parquet; the news family is derived from the text and event corpora and is itself decomposed into a sentiment sub-family and an event sub-family. Six subsections describe the resulting features.

3.6.1 Technical-analysis features

Technical-analysis features describe the structural state of the market at a given timestamp: whether the price is in an uptrend, a downtrend or a range; where the recent supports and resistances lie; and where the most recent swing highs and lows sit (Murphy, 1999). Three groups of features are produced. *Trend-regime* features classify each bar as bullish, bearish or sideways using the Average Directional Index (ADX) together with the slope of a long-period exponential moving average; the regime label gates the rule layer described in Section 3.7. *Support-and-resistance* features track rolling extrema and pivot points: the highest high and lowest low over a configurable window, the previous-day pivot point and its derived support and resistance levels, and the distance of the current price to each of these levels in pip units. *Swing-structure* features identify swing highs and swing lows from the price series via a ZigZag-style detector, exposing the location and magnitude of the last few swings; they form the input to the chart-pattern detection of Section 3.6.3.

3.6.2 Indicator features

Indicator features quantify momentum, volatility and microstructure conditions, building on the standard families surveyed in (Murphy, 1999): trend-following indicators (simple and exponential moving averages, MACD), momentum oscillators (RSI, stochastic oscillator), volatility filters (ATR, rolling realized volatility, Bollinger bandwidth) and microstructure proxies derived from tick data (rolling tick count, average bid–ask spread). Indicator periods follow the conventions familiar to the author from his prior retail Forex trading practice and are fixed before any backtesting, so that hyperparameter optimization does not silently extend over the indicator family. All indicator features are computed with strict point-in-time discipline: at time t , only data with timestamp $\leq t$ enters the feature value, and any rolling-window statistic uses only the trailing window.

3.6.3 Chart and candlestick pattern features

Pattern features encode the graphical-analysis tradition of technical analysis: discrete shapes formed by recent price action that traders have historically associated with directional bias. Two strands are implemented. The first is the Japanese candlestick tradition codified in the West by Nison (1991); the second is the classical chart-pattern tradition cataloged in (Bulkowski, 2005) and given a computational footing by Lo, Mamaysky and Wang (2000).

Candlestick patterns are deterministic functions of one to five consecutive OHLC bars: the Doji, the Hammer, the Engulfing pattern, the Morning and Evening Stars, the Three White Soldiers and the Three Black Crows, among approximately sixty patterns enumerated in (Nison, 1991). The implementation reuses the reference detectors of the open-source TA-Lib library, which exposes one function per pattern (e.g., `CDLDOJI`, `CDLENGULFING`, `CDLHAMMER`). Each detector returns, for every bar, an integer-valued indicator: zero if no pattern is present, a positive

value if a bullish instance is detected, a negative value if a bearish instance is detected. The encoding lets a downstream model treat each pattern as a categorical-with-direction feature without further preprocessing.

Classical chart patterns operate on longer time scales than candlesticks and require the swing-structure features of Section 3.6.1 as input. Patterns considered include head-and-shoulders, double tops and double bottoms, ascending and descending triangles, and bull and bear flags. The detection procedure follows Lo, Mamaysky and Wang (2000): the price series is first smoothed by kernel regression to identify local extrema, the resulting sequence of extrema is matched against parametric pattern templates, and a confidence score is computed from the residual between the observed sequence and the matched template. The pattern catalog and the failure-rate statistics reported by Bulkowski (2005) inform the prior on each pattern’s reliability and the conservative thresholds used to admit a detection into the feature stream. As with all features in this chapter, pattern detections are emitted only at the bar that closes the pattern, so that no future information enters the feature value.

3.6.4 Sentiment features

Sentiment features are produced by two parallel scorers applied to the same news input, in line with the discussion in Section 2.3.3. The first is a dictionary-based baseline using the Loughran–McDonald financial-domain lexicon (Loughran; McDonald, 2011). The second is a transformer-based scorer using the FinBERT family (Araci, 2019; Yang; Uy; Huang, 2020); both FinBERT variants are evaluated and the better-performing of the two on a small in-domain validation set is retained. The transformer scorer takes a tokenized sentence as input and returns a scalar polarity in $[-1, +1]$ together with a confidence score; sentence-level scores are aggregated to article level by length-weighted averaging and to time-bucket level by a confidence-weighted mean over articles in the bucket. Time buckets are aligned to the same minute grid as the price features.

Sentiment is attributed *per currency* rather than per pair. Each article is mapped to the currencies it concerns by combining the currency mentioned with its issuing authority or jurisdiction (ECB and the euro area $\rightarrow$ EUR; the Federal Reserve and the United States $\rightarrow$ USD; the Bank of Japan and Japan $\rightarrow$ JPY), so that a single article may contribute to more than one currency stream and a multi-currency headline with opposing tone is attributed with reduced confidence. The pair-level sentiment feature is then formed as the base-minus-quote polarity differential of the two currency streams (EUR/USD from the EUR and USD streams, USD/JPY from the USD and JPY streams) so that a US-macro release (for example an FOMC decision) propagates to both pairs through the shared USD stream. This mirrors the structure of an exchange rate as the relative valuation of two currencies, in the same spirit as the interest-rate differential of the carry trade, and is the textual analog of the cross-sectional, exposure-based treatment of geopolitical risk discussed in Section 2.4.

The choice to preserve minute-level arrival timing on the sentiment side is not arbitrary.

Hafez and Xie (2013) report that sentiment inflection points extracted from macro news predict intraday EUR/USD moves for several hours after the inflection point, which is exactly the type of temporal structure that would be destroyed by collapsing article arrivals into a daily index before the backtest. The present work therefore preserves article timestamps and only aggregates to the minute bucket required by the price grid.

The dictionary baseline is retained even after the transformer scorer is operational. Tetlock (2007) and Loughran and McDonald (2011) establish that domain-specific dictionaries already provide measurable predictive content in financial settings; the comparison between the dictionary baseline and the transformer scorer therefore acts as a sanity check for the larger model, in line with the methodological caution recommended by López de Prado (2018). Where the inference-latency budget of a future live deployment is binding, the comparison also documents the cost–benefit trade-off between the cheaper lexicon scorer and the more expensive transformer scorer (Section 2.3.3). Whether to escalate to a larger foundation model in the FinGPT (Yang; Liu; Wang, 2023) or BloombergGPT (Wu *et al.*, 2023) class is treated as an open question in Section 3.10 rather than asserted in advance.

Recent LLM-based results in financial sentiment analysis (Fatouros *et al.*, 2023; Ballinari; Maly, 2025; Iacovides; Zhou; Mandic, 2025) motivate that future scorer swap, but they do not justify skipping the FinBERT-class baseline in the present dissertation. Their role here is methodological inspiration for a later perception-layer upgrade, not evidence that the current implementation should bypass the conservative baseline / challenger comparison.

3.6.5 Event features

Event features are derived from GDELT and from the GPR index. From the GDELT Events table the daily counts of CAMEO event categories relevant to Forex (monetary-policy actions, sanctions, military activity, sovereign downgrades) are extracted, together with their average tone and average Goldstein score. From the GDELT Global Knowledge Graph, daily aggregates of the geopolitical-actor co-mention graph centered on the United States, the euro area and Japan are computed. The GPR index of Caldara and Iacoviello (2022) is incorporated as the daily-frequency continuous regime feature. All event features are produced at daily resolution and forward-filled onto the minute grid; the asymmetry in resolution is intentional and reflects the asymmetry in the underlying signal.

3.6.6 Market-activity features

The fifth family captures the relative trading activity of each currency. The foreign-exchange market is over-the-counter and has no consolidated traded volume; the present work therefore uses *tick volume* (the count of price updates per interval, the standard activity proxy in retail Forex platforms) as recorded in the QuoteBar features of Section 3.4. Per-bar tick volume is aggregated across the configured pairs into a daily *currency-strength* signal per currency: the share of total activity attributed to EUR, USD and JPY, together with the day’s activity ranking.

The premise, drawn from the discretionary trader’s heuristic, is that the currency concentrating the day’s activity tends to indicate the direction in which the market is leaning. Because the signal is cross-pair (it cannot be computed from a single pair in isolation) it is materialized at this layer rather than inside the per-pair execution engine. The feature is deliberately universe-aware: with the two pairs of the present work it covers EUR, USD and JPY only partially (USD being shared), and it strengthens as the pair universe is extended toward the major pairs in future work. Like the other price-derived families, it feeds a dedicated context filter in the deterministic chain of Section 3.7.2.

3.7 Fusion and Deterministic Execution

The strategy combines the five feature families of Section 3.6 through a two-phase pipeline. The first phase performs late fusion of the perception layer; the second phase maps the fused output to a trade decision through a deterministic rule implemented as a configurable chain of *filters*. The two phases are designed separately because the methodological commitment of this work hinges on keeping the second phase rule-based even when the first phase incorporates non-deterministic components such as transformer-based language models.

3.7.1 Phase one: late fusion of the perception layer

Five sub-models are trained independently, in line with the late-fusion discussion in Section 2.5. The first four cover the price-derived families (technical-analysis structural features, indicator features, chart-and-candlestick pattern features, and market-activity features). The fifth covers the news-derived family, in which sentiment and event features are concatenated into a single text-derived input matrix. Each sub-model produces a calibrated probability of an upward move at horizon H . A small meta-learner (logistic regression with a single regularization hyperparameter) combines the five probabilities into a single calibrated output $\hat{p}_t \in [0, 1]$.

Late-fusion is the safer default when feature families have different signal-to-noise ratios, as is the case here between price-derived features (low noise, high frequency) and text-derived features (high noise, lower effective frequency). Training all features in a single early-fusion model risks the high-frequency price features dominating the loss surface during gradient descent and squeezing the textual features out of the learned representation.

The choice of sub-model architecture is gradient-boosted decision trees (LightGBM) for each family, on the basis that the feature spaces are mixed-type, moderately sized, and well within the regime where boosting outperforms or matches deep models on tabular finance data. A switch to a small Transformer encoder over the time axis is reserved as future work and is not part of the deliverable for this dissertation.

Two strategy variants are trained on the same data splits: a *baseline* that uses only the four price-derived sub-models (technical-analysis structural, indicator, pattern and market-activity), and a *hybrid* that adds the news-derived sub-model. The two variants share an identical meta-

learner specification; the only difference is whether the news-family probability enters the meta-learner's input. This isolates the marginal contribution of the textual features.

3.7.1.1 Phased delivery from a minimum viable baseline

The architecture above describes the full delivery target. Implementation proceeds in stages, and each stage is itself tradable, so that earlier stages can be empirically validated before complexity is added. The exact family decomposition, the indicator set within each family, and the filter set retained at each stage are deliberately treated as subject to revision based on ablation results: each stage is a checkpoint whose components must demonstrably contribute before the next stage is built on top of it.

The **minimum viable baseline** (Stage 1) covers exactly the components a Forex strategy needs to function and to protect capital: a single price-derived feature set, one LightGBM sub-model producing a calibrated directional probability, a threshold-based entry and exit rule on that probability, fixed-fraction position sizing, an ATR-based stop-loss, and a portfolio-level drawdown cap. This stage proves the data-and-execution pipeline end to end and produces a strategy that can be evaluated for capital growth before any added complexity is justified. Stage 2 splits the price features into per-family sub-models (technical-analysis structural, indicator, pattern, market-activity) and introduces the logistic meta-learner of the previous paragraphs. Stage 3 adds the news-derived sub-model (FinBERT-class sentiment, GDELT event features and the GPR index), defining the hybrid variant of Section 3.7.1. Stage 4 introduces the configurable filter chain of Section 3.7.2 (trend-regime, news-context, risk-guard, capital-management and terminal-threshold filters) on top of the meta-learner output. The progression is monotone in expressiveness: each stage retains the entry-exit-sizing-protection nucleus of Stage 1 and adds optional channels and gates above it.

3.7.2 Phase two: deterministic execution as a filter chain

The trading decision is produced by a configurable chain of deterministic *filters* applied in sequence to the running state. The pattern is borrowed from the layered request-processing model common in Java enterprise frameworks and adopts the FX-industry term *filter* to name the chain elements. Each filter receives the accumulated state, contributes a recommendation in {BUY, SELL, HOLD, NEUTRAL, ABSTAIN} together with a textual reason, and may either pass to the next filter or veto by short-circuiting the chain to HOLD. The complete sequence of filter contributions is persisted as the audit trail for every simulated trade, so each backtest entry is explainable down to the contribution of each filter.

The chain for the present work has eight filters: a trend-regime filter (Section 3.6.1), an indicator filter (Section 3.6.2), a chart-and-candlestick pattern filter (Section 3.6.3), a market-activity filter that reads the daily currency-strength signal (Section 3.6.6) and biases the direction toward the day's dominant currency, a news-context filter that consumes the sentiment and event features (Sections 3.6.4–3.6.5) and vetoes on active high-risk events, a risk-guard filter (draw-

down caps, leverage cap, portfolio-at-risk cap), a capital-management filter that computes the lot size and verifies margin, and a terminal threshold-rule filter that aggregates the accumulated recommendations and emits the final decision. The terminal filter applies the rule

$$\text{decision}_t = \begin{cases} \text{BUY} & \text{if } \hat{p}_t > \theta_{\text{high}} \text{ and } r_t = \text{bull and } v_t = 0, \\ \text{SELL} & \text{if } \hat{p}_t < \theta_{\text{low}} \text{ and } r_t = \text{bear and } v_t = 0, \\ \text{HOLD} & \text{otherwise,} \end{cases}$$

where $\hat{p}_t$ is the meta-learner's probability output from Section 3.7.1, r_t is the trend regime from the trend-regime filter, and v_t is the event-veto flag from the news-context filter. The thresholds θ_{high} and θ_{low} are calibrated on the validation span and held fixed during the test span; they are not learned by the model. Capital management and order execution on top of the decision (position sizing, partial-take-profit ladders, trailing-stop logic, exposure caps per pair) follow a rule-based scheme grounded in standard retail Forex algorithmic-trading practice. The implementation planned for this work is a clean-room rewrite in Python, written from first principles and without reference to any prior third-party codebase, as described in Section 3.9.

Figure 3 makes explicit a distinction that the prose description alone can obscure. The chain is drawn in layers organized by function rather than by exact implementation order, and not every filter plays the same role. The top layer contains context and confirmation filters that accumulate directional or situational evidence from price structure, indicators, patterns and news context; the middle layer contains hard veto gates on execution viability, risk and capital constraints, any one of which short-circuits the chain to HOLD; the bottom layer is the terminal rule that maps the surviving state to BUY, SELL or HOLD. The nodes marked $\dots$ indicate that the chain is extensible and may include additional filters not shown explicitly: some gates in the diagram are illustrative and need not all be present in the first implementation. The purpose of the figure is to make explicit the distinction between evidence-contributing filters and veto-capable execution gates.

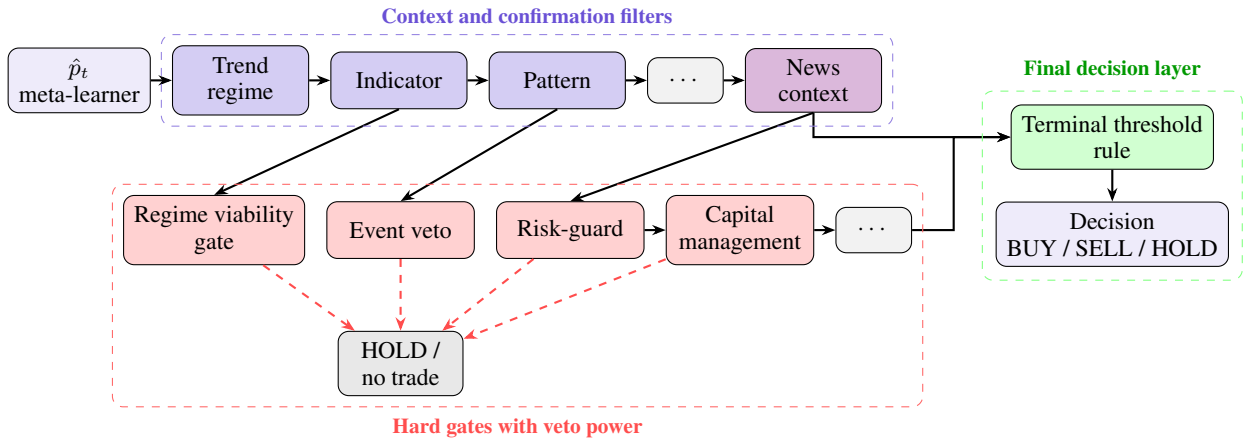


Figure 3 – Deterministic execution chain.

The chain is configurable: the order, presence and parameters of each filter live in a YAML configuration file rather than in the engine code, so adding, removing or reordering filters in subsequent strategy variants requires no engine changes. This is the structural device that makes the future-work tiers of Chapter 5 feasible without re-architecting the backtester: an additional perception agent or a consensus aggregator slots in as another filter in the chain.

The *order* of the filters is deliberately treated as a configurable design choice rather than a fixed assumption. Because each filter acts on the state accumulated by the filters before it, two orderings of the same set of filters can produce different decisions, and there is no a priori reason to commit to one ordering in advance. The present work therefore determines the filter ordering empirically: candidate orderings, together with the presence or absence of individual filters, are evaluated by simulation, and the configuration that performs best is retained. To keep this search consistent with the validation protocol of Section 3.8, the ordering is treated as a hyperparameter of the execution layer rather than as a free parameter of the final evaluation. It is selected on the training and validation folds of the combinatorial purged k -fold scheme, and the chosen ordering is frozen before the single held-out test pass. The flexibility is thus a search space explored only on in-sample data, never a degree of freedom exercised on the test span, so it cannot inflate the reported out-of-sample result. The comparison between competing orderings is reported as part of the ablation study in Chapter 4 (Section 4.5).

The same treatment applies to the *timeframe* on which each filter operates. The strategy is multi-timeframe: a decision is taken on the finest bar of the trading grid, but an individual filter may analyze a coarser timeframe to contribute its evidence about that decision, in the manner of a discretionary trader who reads a higher timeframe for context and a lower one for entry. A trend-regime filter may therefore evaluate, say, the four-hour bar while the entry decision is made on the one-minute bar. The supported timeframes are the standard Forex set (one, five, fifteen and thirty minutes; one and four hours; one day), and each filter declares the timeframe it reads in the same configuration file as the chain order. Because every reusable bar series is precomputed once by the transformation stage of Section 3.4 and read through a cache (Section 3.3), making a filter observe a different timeframe costs a lookup, not a recomputation: the canonical Parquet store holds each timeframe, materialized ahead of the simulation sweep, and any timeframe not yet present is computed once on first request and reused thereafter. As with the chain order, the per-filter timeframe is a hyperparameter selected on the training and validation folds and frozen before the held-out test pass; the choice of which timeframe each layer reads is therefore part of the same ablation, not a free parameter of the final evaluation.

3.7.3 Extended filter set: perception and execution quality gates

The filter chain described above (F1–F7) forms the baseline. The review of prior monographs in Chapter 2 suggests seven additional filters that can be configured into the chain without structural changes. Each one is motivated by a specific gap or methodological caution identified in the related work.

3.7.3.0.1 F8: Candlestick Confirmation Gate (Felix, 2024).

Blocks a trade if no confirmatory candlestick pattern appears in the trailing k bars. The motivation comes from Felix (2024), which explicitly encodes candlestick patterns as input variables in an ML swing-trading study. The present filter is therefore an adaptation of that modeling choice, not a direct reproduction of Felix’s full feature set: it uses a deterministic confirmation gate before the risk layers. The initial catalog prioritizes patterns explicitly discussed in the monograph (Hammer, Shooting Star, Bullish/Bearish Engulfing, Harami, Doji, Inverted Hammer, Three Black Crows and Morning Star) and can later be extended through the TA-Lib library (CDL* functions), which provides standardized, deterministic pattern recognition based on the Nison (1991) and Bulkowski (2005) specifications. Parameters: $k = 5$ (configurable), pattern list (expansível até ~60 padrões do TA-Lib). Position in chain: between F4 (news-context) and F5 (risk-guard), as a final price-confirmation check before risk gates.

3.7.3.0.2 F9: Regime Viability Gate (Hô Don Lee, 2014).

Implements a *regime-viability* notion that combines the practical trading principle of not forcing trades in structurally poor conditions (Miner, 2009) with the *tradability* concept of Lee (2014): a setup may appear statistically admissible yet still not be operationally tradable if volatility is excessive, direction is absent, or the local state is unstable. The point of contact between the two references is the distinction between the existence of a signal and the existence of a market state in which executing that signal is justified. The present work adopts that conceptual distinction but not Lee’s implementation details, which are too fragile for direct reuse: the original monograph relies on an Excel implementation, a 63-day sample, full Bloomberg dependence and a non-standard Tikhonov–Miller regularization step. The filter therefore re-specifies the idea in a cleaner and more scalable form through three sub-tests: (i) ATR within band ($\pm k$ standard deviations), (ii) ADF stationarity test on recent closes, (iii) sentiment polarity stability (fewer than m sign flips in trailing window). Parameters: $k = 2.0$, $\alpha_{\text{ADF}} = 0.05$, $m = 3$, lookback $n = 60$ bars. Position in chain: first gate of Phase 2; if the regime is not viable, no further filters are evaluated. **Note:** F9 differs from F1 (trend-regime filter) in scope: F1 classifies trend direction (bull/bear/sideways), while F9 assesses whether the current regime is *operationally tradable* regardless of direction. A market can be in a clear uptrend (F1 = bull) yet still untradable due to excessive volatility or instability (F9 = veto).

3.7.3.0.3 F10: Textual Confidence Gate (Ferrarini, 2025).

Suppresses the news sub-model probability when the FinBERT scorer is uncertain. This gate is an engineering extension inspired by Ferrarini (2025), who calibrates textual classifiers with ROC analysis and Youden’s J rather than accepting raw scores uncritically. Here that same calibration discipline is translated into a simple confidence rule based on the softmax

margin (difference between top-2 class probabilities). If $\text{margin} < \theta_{\text{conf}}$, the news probability is neutralized to 0.5 (no directional signal). Parameters: $\theta_{\text{conf}} \in [0.15, 0.30]$. Position in chain: Phase 1, between FinBERT scorer and news sub-model.

3.7.3.0.4 F11: Cross-Model Agreement Gate (Ferrarini, 2025).

Reduces the weight of the news sub-model when FinBERT and the Loughran–McDonald dictionary scorer disagree strongly. This is likewise an adaptation inspired by Ferrarini (2025): Ferrarini compares multiple textual models and explicitly suggests combined pipelines as future work, whereas the present dissertation translates that idea into a conservative agreement gate for financial news. If both models exhibit high confidence ($|s| > \theta_{\text{mag}}$) and opposite signs, the news weight multiplier is reduced to 0.5 (or 0.0 in a stricter variant). Parameters: $\theta_{\text{mag}} \in [0.30, 0.50]$. Position in chain: Phase 1, after F10.

3.7.3.0.5 F12: Sub-Model Consensus Gate (Veiga, 2022).

Requires that at least m of the 4 sub-models agree on direction. The gate is not claimed as a method proposed by Veiga (2022); it is a conservative extension motivated by Veiga’s explicit warning that optimization on stochastic trading processes is highly prone to overfitting. In the present work, that warning is operationalized by demanding minimum directional agreement across independently trained sub-models before execution is allowed. Each sub-model produces a calibrated probability; the gate counts BUY votes ($p > 0.55$), SELL votes ($p < 0.45$), and NEUTRAL otherwise. If $\max(\text{BUY}, \text{SELL}) < m$, the trade is vetoed. Parameters: $m \in \{2, 3, 4\}$, neutral band $[0.45, 0.55]$. Position in chain: Phase 2, after meta-learner, before trend-regime filter.

3.7.3.0.6 F13: Duplicate Detection Gate (Pugina, 2024).

Discards news articles that are near-duplicates of recently processed items (syndicated wire stories). Pugina (2024) does not propose duplicate filtering; the contribution borrowed from that monograph is narrower and explicit: SBERT embeddings can be used as a practical text representation in a downstream decision pipeline. The present dissertation reuses that representation for a different purpose, namely cosine-similarity deduplication over the trailing window of recent headlines. If $\max(\text{sim}) > \theta_{\text{dup}}$ against the trailing w embeddings, the article is skipped. Parameters: $\theta_{\text{dup}} = 0.95$, $w = 100$. Position in chain: Phase 1, before any sentiment scorer.

3.7.3.0.7 F14: Entity Relevance Gate (Brito, 2025).

Filters news that fails to mention entities relevant to the currency pair (ECB, Fed, EUR, USD for EUR/USD; BOJ, JPY for USD/JPY). Brito (2025) does not define pair-specific FX filters; its relevant contribution is the broader idea of using a semantic mediation layer to decide

what information should reach later stages of processing. The present work adopts a lightweight version of that intuition through a simple keyword/entity gate with a predefined entity set and configurable minimum match count. Parameters: `min_matches = 2`. Position in chain: Phase 1, early in the ingestion pipeline.

3.7.3.0.8 F17: Cross-Market Risk-Off Gate (Laïdi, 2009; Chan, 2013).

Blocks long USD/JPY (carry-direction) entries when cross-asset state signals risk-off conditions. The gate is motivated by the carry-trade unwind mechanism documented in Laïdi (2009) (Ch. 5, pp. 115–126): structural current-account asymmetry makes JPY a funding currency, so volatility-of-volatility shocks force unwinding flows that move USD/JPY sharply against the carry direction. Chan (2013b) (p. 184) provides a concrete VIX threshold: mean-reversion performance on the FSTX index dropped from 13 % APR to 2.6 % APR once $VIX > 35$. The filter combines three sub-checks in disjunction: (i) $VIX_t > 35$, (ii) rolling 20-day gold–USD correlation outside $[-1, -0.4]$ (anomalous regime relative to the structural $\rho \approx -0.84$ reported by Laïdi, p. 7), or (iii) entry timestamp inside the 21:00–01:00 UTC yen-unwind window (Laïdi, p. 123). Position in chain: Phase 2, after F9.

3.7.3.0.9 F18: S&P Alignment Gate (Laïdi, 2009; USD/JPY only).

Laïdi (2009) (Fig. 5.5, p. 120) documents a structurally positive correlation between USD/JPY and the S&P 500 that emerged when the Fed began raising rates in June 2004 and persists through the present TCC window. The filter blocks entries whose proposed USD/JPY direction is contradicted by the previous day’s S&P 500 return by more than 1.5σ . The gate is *pair-specific*: it does not apply to EUR/USD, where the equity correlation is structurally weaker and not direction-stable. Position in chain: Phase 2, USD/JPY-only branch of the chain.

3.7.3.0.10 F19: Positioning Extreme Veto (Saettele, 2008).

Adds a hard-data sentiment channel that is orthogonal to the FinBERT-derived NLP signal. Saettele (2008) (Ch. 5, pp. 83–91) constructs three positioning indices from the CFTC Commitments-of-Traders report (a Composite COT Index, a speculative %Long Ratio Index and a commercial %Long Ratio Index, all rank-transformed by percentile over a 52-week window) and reports that simultaneous alignment of the three indices at the 0 or 100 extremes coincided with major USD-index turns in 1989, 1992, 2001 and 2004. The present filter operationalizes that triple-alignment rule: if all three indices are ≥ 95 (or ≤ 5) and point against the proposed entry, the trade is vetoed. The convention of Saettele (p. 76) is respected for USD/JPY: extreme bullish COT on JPY futures (CME) corresponds to a bearish signal on the USD/JPY spot. The filter adds a textual-independent sentiment channel; the present work does not adopt Saettele’s keyword-based headline taxonomy (Ch. 4), which has been superseded by contextual scoring with FinBERT-class models. Position in chain: Phase 2, after F12.

3.7.3.0.11 F20: Data-Mining Significance Gate (Aronson, 2007).

A configuration-time gate, not a runtime gate. Before any candidate filter combination is promoted from the validation span to the held-out test span, its observed Sharpe ratio must exceed the 95th percentile of the Monte Carlo Permutation Test distribution defined in Section 3.10.1. The gate is configuration-level rather than execution-level because it acts on the full backtest history of a candidate variant rather than on per-bar state. It implements the protective wrapper that Aronson (2007) (Ch. 6, pp. 327–329) prescribes for any rule universe whose members share components and are therefore correlated, precisely the case of the F1–F19 chain. Parameters: $M = 1,000$ permutations, $\alpha = 0.05$. Position: offline, between validation-span training and test-span evaluation.

Implementation parameters (excerpt from configuration YAML).

```
filters_phase1:
  f10_confidence: {enabled: true, threshold: 0.20}
  f11_agreement:  {enabled: true, magnitude_threshold: 0.30}
  f13_dedup:      {enabled: true, similarity_threshold: 0.95, window: 100}
  f14_entity:     {enabled: true, min_matches: 2}
filters_phase2:
  f9_regime_viability: {enabled: true, atr_k: 2.0, adf_alpha: 0.05}
  f8_candlestick:      {enabled: true, lookback: 5}
  f12_consensus:       {enabled: true, min_agreement: 3}
```

Effort prioritization. F10, F11 and F14 are *zero-dependency* additions (45 lines, no new packages) and can be implemented in a single session. F8 and F12 require TA-Lib functions already present. F9 requires `statsmodels` for the ADF test. F13 requires `scikit-learn` for cosine similarity and `sentence-transformers` for SBERT encoding. The recommended implementation order is: F10 → F11 → F14 → F8 → F12 → F9 → F13.

3.7.4 Strategy variants for ablation

Six strategy configurations are planned for the ablation stage of Chapter 4:

- **Core:** F1–F7 only (baseline).
- **Variant A:** Core + F8. Tests the marginal contribution of candlestick confirmation.
- **Variant B:** Core + F10 + F11. Tests quality gating on the textual perception layer.
- **Variant C:** F9 + F1–F7 + F12. Tests regime viability and sub-model consensus.
- **Variant D:** All quality gates (F9, F10, F11, F13, F14 in Phase 1; F1–F8, F12 in Phase 2). Tests full-stack quality gating (risk: overfiltering).

- **Variante E:** Variante D plus les cross-actifs et les positions (F17, F18 sur USD/JPY uniquement, F19). Teste la contribution marginale des canaux de sentiment de marché et de données dures qui sont orthogonaux au signal dérivé de FinBERT.

Chaque variante est entraînée et évaluée sous le même protocole CPCV, permettant une attribution claire des différences de performance au jeu de filtres. Toutes les six variantes sont soumises au seuil de Signification du Data-Mining (Section 3.7.3) avant promotion de la validation à l'ensemble de test; le nombre de six variantes est inférieur à la limite de Bailey-Borwein-López-de-Prado discutée dans la Section 3.10.1.

Cette séparation entre la couche de perception non-déterministe et la chaîne de filtres déterministe est l'engagement méthodologique central de l'ouvrage. La couche de perception est autorisée à être non-déterministe (un modèle BERT finement ajusté produit des scores de sentiment, et un modèle de langage plus avancé pourrait à l'avenir le remplacer pour une extraction de caractéristiques plus riche) car ses sorties sont calculées une fois et lues déterministiquement au moment du backtest, via le cache de lecture et le stockage d'exécution matérialisés de la Section 3.3, et ne sont jamais recalculées dans la boucle de replay. La chaîne de filtres, au contraire, est basée sur des règles, correcte en temps réel, et entièrement reproductible. Avec les mêmes caractéristiques mises en cache et les mêmes seuils calibrés, l'ensemble du backtest est reproductible quelle que soit la non-déterminisme de la couche de perception en amont. Le schéma est donc celui de *perception agentic, exécution déterministe*: la couche de perception peut intégrer des modèles de plus en plus performants sans que la chaîne de décision perde sa traçabilité, sa reproductibilité ou ses garanties de validité statistique. Le chemin de migration vers des couches de perception plus performantes est enregistré dans la conclusion comme travail futur.

3.8 Backtesting and Evaluation Protocol

Le protocole d'évaluation suit la discipline méthodologique établie dans (López de Prado, 2018). Trois engagements sont pris avant que tout modèle ne soit entraîné.

Premier, l'entraînement *walk-forward*. Les données sont divisées chronologiquement: 2015 à 2022 forment l'ensemble d'entraînement, 2023 l'ensemble de validation, et 2024 l'ensemble de test (soumis à la règle de la Section 3.5.3 qui pourrait resserrer la limite inférieure). Dans l'ensemble d'entraînement, les modèles sont réentraînés de manière itérative: à chaque point de réentraînement, la fenêtre d'entraînement se termine, une nouvelle tranche de validation est isolée, et le modèle est réentraîné sur les données strictement antérieures à la tranche. Aucune information future n'est autorisée à entrer dans les valeurs des caractéristiques ou les paramètres du modèle à n'importe quel instant.

Deuxième, la validation croisée purgée combinatoire. Dans chaque fenêtre d'entraînement, les plis de validation sont construits à l'aide du schéma *k*-pli purgé combinatoire de López de Prado (2018): les observations dont l'horizon de label chevauche n'importe quel pli de validation sont purgées de l'ensemble d'entraînement, et une période d'embargo supplémentaire après chaque pli de validation est exclue. Cela traite la fuite d'information qui survient lorsque la variable dépendante est calculée sur un horizon avant.

Troisième, la transparence des rapports. Chaque résultat rapporté identifie l'étendue des données, le schéma de validation et les métriques. Les métriques sont: le ratio de Sharpe annualisé (après soustraction d'un explicite

transaction-cost model), the maximum drawdown, the hit rate (proportion of trades closed in profit), the average holding time, and the turnover. The Sharpe ratio alone is too easily inflated by selection bias on short horizons (López de Prado, 2018); the supplementary metrics constrain the interpretation of any reported headline number.

The transaction-cost model is empirical. From the tick Parquet, the rolling distribution of bid–ask spreads is computed for each pair. Each simulated trade incurs a half-spread cost on entry and on exit, plus a fixed slippage term calibrated against the larger end of the realized-spread distribution. Commissions are absorbed into the spread for the demo-account regime under which the back-tested data are obtained. The same cost model applies to the baseline and to the hybrid strategy, so that the comparison reports the marginal contribution of textual features net of identical execution friction.

3.8.1 Numbered Experiments

The empirical evaluation reported in Chapter 4 is organized as five numbered experiments. Each one has an explicit target, a key manipulated variable, and a primary metric. Table 4 summarizes the plan; each row maps to one section of Chapter 4.

Table 4 – Numbered experiments. Each row maps to one section of Chapter 4.

#	Target	Filters activated	Key variable	Primary metric
1	Calibrate price-only baseline on EUR/USD	F1–F7 (baseline)	LightGBM hyperparameters	Sharpe, max draw-down
2	Evaluate full hybrid vs baseline on EUR/USD	F1–F7 (+ news sub-model)	Presence/absence of news	Sharpe, drawdown, hit rate
3	Replicate Experiment 2 on USD/JPY	F1–F7 (+ news)	Currency pair	Same metrics + stat. test
4	Ablation: feature family contributions	F1–F7 (varying sub-sets)	News sub-family activated	Same metrics
5	Compare vs Zhang (2025)	F1–F7 (+ news)	Validation protocol	Sharpe ratio
6	Filter ablation: Core vs A/B/C/D	Core, A (+F8), B (+F10/11), C (+F9/12), D (all)	Extended filter set F8–F14	Sharpe, drawdown, turnover
7	F9 sensitivity analysis	F9 only	k_{ATR} , α_{ADF}	Trades/year, win rate
8	F10/F11 threshold sweep	F10, F11	θ_{conf} , θ_{mag}	News contribution
9	F12 consensus test	F12 only	$m \in \{2, 3, 4\}$	False positive rate

The sequencing is intentional. Experiment 1 calibrates the baseline before any text-derived feature is involved. Experiment 2 introduces the news sub-model and measures the marginal contribution that motivates hypothesis H1 of Section 1.1. Experiments 3 and 4 test robustness across pair and feature sub-family. Experiment 5 documents the protocol divergence from Zhang (2025).

Experiments 6–9 focus specifically on the extended filter set of Section 3.7.3. Experiment 6 conducts the main ablation study comparing the Core configuration (F1–F7 only) against the four strategy variants (A, B, C, D) defined in Section 3.7.4, isolating the marginal contribution of each filter subset. Experiments 7–9 perform sensitivity analyses on individual filters (F9, F10/F11, F12) to calibrate their parameters without the confounding effects of other filter interactions. All non-deterministic metrics are reported as mean $\pm$ standard deviation over the combinatorial purged k -fold folds, never as single-run values.

3.9 Execution Engine

The backtests of the present work target the open-source LEAN engine maintained by QuantConnect⁶. The portion of the integration runnable at the time of submission is reported in Chapter 4. Three properties drive the choice. First, LEAN is published under the Apache 2.0 license, which imposes no copyleft obligations on derivative work, a property that an eventual commercial deployment of the strategy can rely on. Second, an official OANDA v20 brokerage adapter⁷, also Apache 2.0, is available; the same engine therefore drives both the backtest reported in this work and any future paper-trading experiment. Third, LEAN exposes a Python algorithm interface, so strategy code can be expressed in the same language as the rest of the data pipeline. Should a hot path become Python-bound in future work, a Rust crate compiled via PyO3 can be loaded as a Python extension module and called from the strategy without changes to LEAN itself.

A clarification on cost is warranted, since the LEAN name is often associated with paid usage. That association concerns the QuantConnect *cloud* platform (its hosted backtest and live-trading compute nodes, and its subscription Data Library) and not the engine. The present work uses neither: it runs the open-source LEAN engine *locally*, through the LEAN command-line interface inside a Docker container, and feeds it data the project downloads and prepares itself (Section 3.4). No QuantConnect cloud subscription, compute node or hosted dataset is involved, so backtesting, including the parameter and ablation sweeps that dominate the experimental work, incurs no platform cost beyond local compute. This is also why the data architecture of Section 3.3 is built around a local, bulk-downloaded store rather than a hosted data feed. An eventual live deployment could either remain self-hosted, against the OANDA adapter, or adopt a paid cloud node; that decision lies outside the scope of the present work.

The alternatives considered and rejected illustrate the trade-offs. Backtrader is a mature Python backtester but is licensed under GPL v3, the viral copyleft of which is incompatible with future commercial reuse. NautilusTrader offers a modern Python/Rust architecture but lacks an OANDA brokerage adapter and is licensed under LGPL v3; both factors raise integration cost. Vectorbt is permissively licensed but provides only a research-style backtester without a

⁶ <https://github.com/QuantConnect/Lean>

⁷ <https://github.com/QuantConnect/Lean.Brokerages.Oanda>

live-execution layer. LEAN is the option that satisfies all of: permissive license, official Forex brokerage support, Python algorithm language, and runs natively on Linux.

The data are made available to LEAN through the durable `lean-data/` execution store discussed in Section 3.3. A Parquet-to-LEAN materializer performs the conversion: it writes a minute-resolution copy of the price Parquet, in the file layout LEAN’s native data path expects, once and then reuses it across every run of a sweep; an optional `tmpfs` copy at `/dev/shm/lean-data/` may stage it in main memory for faster reads. The materializer’s minute write path is implemented and pinned to a fixed engine image (`quantconnect/lean:17748`); an automated integration test runs that exact image in a container and confirms that bars written by the materializer are read back by the engine at their original instants. That test settled a convention the project now relies on, and which is not obvious from the documentation: LEAN stores Forex (OANDA) minute data in Coordinated Universal Time, indexed by each bar’s start, so the canonical UTC Parquet maps one-to-one onto the engine’s bars with no timezone offset. The materializer is a read-through, one-time tool per ingestion update: it runs on the first cache miss, takes a few seconds, produces on the order of one hundred megabytes per pair at minute resolution, and is not re-run per backtest. Sentiment and event features are exposed to the strategy through LEAN’s `PythonData` extension mechanism; consistently with the read-through model of Section 3.3, they are precomputed once into the consolidated decision features and read from the materialized store rather than recomputed on each run, so no model inference and no repeated transformation occur on the replay path.

The capital-management and order-execution logic that sits on top of the deterministic decision layer (Section 3.7.2) is planned as a clean-room implementation in Python: the code is to be written from scratch for this work, drawing on standard retail Forex algorithmic-trading practice, and without reference to any prior third-party codebase. The rules (Average-True-Range-based position sizing, partial-take-profit ladders, trailing-stop logic and exposure caps per pair) are standard constructions in retail Forex algorithmic trading and are to be implemented as Python modules compatible with the LEAN algorithm interface. The implementation status of each rule at the time of submission is reported in Chapter 4.

3.10 Reproducibility and Threats to Validity

The present section organizes threats to validity along the four-bias taxonomy of Halls-Moore (2015): optimization bias, look-ahead bias, survivorship bias and cognitive bias. To these are added two cross-cutting risks specific to the present pipeline, data coverage drift and process-related scope creep, discussed at the end of the section.

The principal data-related risk is coverage drift: a corpus that is densely populated in 2018 may thin out by 2024 or vice versa, distorting the time-series properties of the resulting sentiment series. The coverage rule of Section 3.5.3 addresses this by deriving the training window from observed data rather than asserting it *ex ante*.

A secondary data-related risk is look-ahead leakage from headline timestamps. The

GDELT and CC-News timestamps reflect publication time but the underlying article may report events from earlier, and aggressive timestamp harmonization can therefore introduce subtle leakage. More importantly, the publication timestamp alone is not the moment at which the corresponding textual feature is operationally available to a trading model: collection, scoring (the FinBERT inference, the Loughran–McDonald pass, the GDELT event-coding pipeline), aggregation into bucket-level features and propagation to the decision layer all add latency, none of which is captured by the bare publication timestamp. A backtest that consumes a feature at its publication minute therefore reads it earlier than any realistic live deployment could, biasing measured performance upward. The procedure adopted is conservative: every textual feature carries a fixed **publication-to-use delay of ten minutes**, that is, a news item published at minute t becomes available to the model only from minute $t + 10$ onward. The same rule applies to GDELT event aggregates: the daily aggregate that closes at the end of day d is available to the model only from the start of day $d + 1$. The ten-minute setting is deliberately conservative: it is large enough to absorb realistic delays in network fetching, inference, feature aggregation and read-through caching, and small enough that the intraday signal of news arrival is not erased. The conservative bias is preferred to a more aggressive setting because it favours reproducibility of the backtest in a live deployment over optimistic in-sample performance.

The principal model-related risk is backtest overfitting in the sense of López de Prado (2018) and Aronson (2007): large hyperparameter searches over the same data tend to produce in-sample Sharpe ratios that are not informative about out-of-sample performance. Aronson (2007) demonstrated this empirically by testing 6,400 binary rules on the S&P 500 and showing that data-mined rules exhibit upwardly biased historical performance. The protocol therefore restricts hyperparameter search to a small explicit grid per sub-model and reserves the 2024 test span for a single evaluation pass per strategy variant. No iteration on the test span is permitted after the first pass.

3.10.1 Rule-level significance under multiple comparisons

Cross-validation alone is not sufficient when a universe of candidate filter combinations is screened against the same data. Aronson (2007) formalises this as data-mining bias, defined as “the expected difference between the observed performance of the best rule and its expected performance” (Ch. 6, p. 256), and shows that the magnitude of the bias scales with the number of rules tested, the rule-correlation structure and the heavy-tailedness of the return distribution (pp. 295–319). Two corrections are applied in the present work.

First, a Monte Carlo Permutation Test in the form of Aronson (2007) (pp. 327–329) is run on each candidate strategy variant. Rule output values (the meta-learner direction at each timestamp) are randomly re-paired with scrambled future market returns without replacement, preserving the rules’ correlation structure with each other while breaking the rule–return relationship. The procedure is repeated $M = 1,000$ times to construct the permutation distribution of the mean rule return under the null hypothesis that the rule’s output is randomly correlated

with the next-period return. A configuration is reported as significant only if its observed Sharpe ratio exceeds the 95th percentile of the permutation distribution. This test is preferred to White (2000)’s Reality Check in the present setting because the filter chain produces highly correlated rule variants, and the permutation procedure preserves that correlation structure where the bootstrap of White does not.

Second, the headline Sharpe ratios of Chapter 4 are reported alongside their deflated counterparts in the sense of Bailey *et al.* (2017). The deflated Sharpe ratio adjusts the observed Sharpe for the number of independent trials performed and for the non-Gaussianity of the return distribution. The same paper provides a minimum-backtest-length bound that constrains how many variants may legitimately be screened within a fixed evaluation window before false discovery dominates: at approximately ten years of daily data, on the order of sixty independent strategy variants is the upper end of what may be tested without dominant overfitting risk. The ablation set of Section 3.7.4 is well below that bound; the extended filter ablations of experiments 6–9 are designed to remain within it.

A useful calibration reference for what counts as a defensible Sharpe ratio in an LLM-augmented setting under realistic costs is the recent *FinDPO* study of Iacovides, Zhou and Mandic (2025), which reports an annual return of 67 % and a Sharpe ratio of 2.0 under a transaction-cost assumption of 5 basis points on long–short portfolios built from S&P 500 equities. Because the evaluation is on equities rather than Forex, the figure should be read as a calibration anchor rather than as a direct comparable for the present work’s Forex evaluation. Read in that role, it bounds the plausible-claim region: a deflated Sharpe materially above 2.0 on a fresh out-of-sample window, after the costs of Section 3.8, should be treated with the scrutiny that López de Prado (2018) prescribes.

The defensive discipline of this section (combinatorial purged k -fold cross-validation, embargoing, frozen filter thresholds, single-pass test-span evaluation, Monte Carlo Permutation Testing and deflated-Sharpe reporting) is selected explicitly against the ten failure modes cataloged in Prado (2018), who reviews the common pitfalls under which quantitative funds applying machine-learning techniques deliver in-sample backtests that fail to survive live deployment. Six of the ten pitfalls are directly addressed by elements of the present protocol: *Pitfall #2 (Research Through Backtesting)* by the frozen-threshold rule and the single-pass test-span evaluation of Section 3.8; *Pitfall #3 (Chronological Sampling)* by the walk-forward training schedule; *Pitfall #7 (Weighting of Non-IID Samples)* by the purging-and-embargoing scheme; *Pitfall #8 (Cross-Validation Leakage)* by the combinatorial purged k -fold construction; *Pitfall #9 (Walk-Forward (Historical) Backtesting)* by the combinatorial extension of walk-forward that Prado himself proposes as the remedy; and *Pitfall #10 (Backtest Overfitting)* by the deflated-Sharpe reporting and Monte Carlo Permutation Test of Section 3.10.1. The present protocol does not claim to neutralize those failure modes; it claims only to apply each documented remedy and to make its residual effect on the headline metrics legible to the reader.

The principal process-related risk is scope creep relative to the available time budget.

The defense date for the present work is set by the University of São Paulo at São Carlos and is treated as the manuscript-freeze deadline. The build is therefore sequenced as a layered minimum viable product (MVP), in three tiers.

The inner tier covers the price ingestion, the deterministic filter chain and the LEAN backtest on EUR/USD. It is non-negotiable. The second tier covers the FinBERT scorer with GDELT and the GPR regime feature. It converts the MVP from a price-only baseline into the hybrid strategy whose evaluation is the subject of Chapter 4. The outer tier covers the additional corpora discussed in Section 3.5: FNSPID, CC-News, the central-bank releases and the Reddit Pushshift archive. The outer tier improves the hybrid without changing its architecture.

The sequencing is graceful. If the manuscript freeze arrives before an outer-tier item has been integrated, the corresponding section of the experimental chapter reports the work as documented future work rather than as silent omission. Furthermore, the unit-of-work abstraction in Section 3.5 guarantees that adding a deferred adapter post-defense does not invalidate the present results. The advantage of this layered approach is that the contribution of the work is preserved at every point along the schedule. A disadvantage is that the outer-tier items, which carry the most novelty in source coverage, are also the items most likely to be deferred.

A residual reproducibility concern is the closed access of some text sources. CC-News and FNSPID are openly redistributable; the Reddit Pushshift dumps remain available as historical archives; the Trump Twitter Archive is publicly maintained. The central-bank corpora are scraped from public web archives at fixed URLs. Together these sources provide a fully re-runnable text-data pipeline.

Determinism, seeds and library pins

Every non-deterministic library used in the present work is seeded explicitly. The seeds are committed to source control and quoted here for auditability:

- `numpy.random.seed(42)`
- **LightGBM:** `random_state=42` on every `Booster`
- **scikit-learn:** `random_state=42` on every estimator that exposes the argument
- **PyTorch (for FinBERT inference):** `torch.manual_seed(42); torch.use_deterministic_algorithms(True)` where supported by the CUDA back-end

Library versions are pinned in `requirements.txt`; the exact pins in force at the time of submission are reproduced in Chapter 4 alongside the experimental results. The hardware environment is documented in the same chapter (CPU model, GPU presence and driver version), since FinBERT inference times depend on it.

Reporting non-deterministic metrics. Every metric that depends on a model fit is reported as mean $\pm$ standard deviation over the combinatorial purged k -fold folds, never as a single-run

value. The convention applies to Sharpe ratio, maximum drawdown, hit rate, average holding time and turnover, and to any classification or regression diagnostic that appears in Chapter 4.

3.11 Final Remarks

The methodology described in this chapter is engineering-led: every architectural choice (Parquet as the canonical layer, a durable `lean-data/` execution store materialized once for replay, late fusion of feature families, walk-forward training with combinatorial purged cross-validation, LEAN as the execution engine) is motivated by a stated constraint, and each open question (the FinBERT versus larger-foundation-model trade-off, the inclusion of optional corpora, the eventual transition to a transformer-based fusion model) is recorded as future work rather than assumed away. The next chapter reports the experimental evaluation that this pipeline produces.

4 EXPERIMENTAL EVALUATION

Work in progress. The present version of this manuscript is an intermediate submission. The experimental chapter is being completed iteratively as the engineering components described in Chapter 3 are brought to a runnable state. The final version of this document, submitted for the defense, will report the results enumerated below.

The chapter is planned in seven sections. Section 4.1 (planned) will describe the implementation status at submission time (which adapters, which scorers, which strategies are runnable end to end) and the exact data spans used. Section 4.2 (planned) will report the coverage matrix of news corpora over the target window and the empirical training-span derived from the coverage rule of Section 3.5.3. Section 4.3 (planned) will report the price-only baseline on EUR/USD (and on USD/JPY where time permits) under the protocol of Section 3.8: Sharpe, maximum drawdown, hit rate, average holding time and turnover. Section 4.4 (planned) will report the hybrid strategy under the same protocol and the same metrics. Section 4.5 (planned) will report the ablation between price-only, price-plus-sentiment, price-plus-events and full hybrid. Section 4.6 (planned) will report a side-by-side comparison against Zhang (2025) on the metrics in common (Sharpe, drawdown), with explicit discussion of why any divergence in headline Sharpe is expected from the stricter validation protocol of Section 3.8. Section 4.7 (planned) will close the chapter with the threats-to-validity discussion specialized to the actually observed results.

Items deferred at submission time, if any, will be listed in the experimental chapter as documented future work rather than as silent omissions, consistent with the layered MVP commitment of Section 3.10.

4.1 Experimental Setup

This intermediate version reports the implementation status of the experimental pipeline at the time of submission, as anticipated in Section 3.4. The engineering work is being delivered in vertical layers rather than all at once, and the present submission documents which layers are runnable end to end and which remain in progress, so that the deferral of empirical results is explicit rather than silent (Section 3.10).

Two components are implemented and pass the project’s quality gate. The first is a shared core library holding the cross-cutting contracts of the pipeline: the instrument abstraction that carries the LEAN-aligned system identity of Section 3.9 (symbol, security type, market and the pip convention that distinguishes the yen-quoted pair), the canonical storage layout of Section 3.3 (the partitioned Parquet tree and the derived execution store), the configuration policy, the read-through cache port, and the repository abstraction over the Parquet store. The second is the Forex price-ingestion stage of Section 3.4, realized as a self-registering Dukascopy adapter that retrieves the broker’s native tick payloads and writes them unmodified to the raw store, leaving

normalization to the downstream transform stage. Both components are developed test-first under a behavior-driven regime and, at submission, comprise more than seventy automated tests at ninety-nine per cent line coverage, passing the static-analysis and strict type-checking gate.

The remaining stages are specified in Chapter 3 but not yet runnable: the transform stage that resamples raw ticks into the minute-resolution `QuoteBar` Parquet of Section 3.4; the sentiment and event feature layer of Section 3.6; the late-fusion model of Section 3.7; the LEAN-based execution engine of Section 3.9, whose `Parquet-to-lean-data` materializer and timezone convention are nonetheless already implemented and validated by a containerized integration test against the pinned engine image, leaving the strategy algorithm and the backtest run as the remaining work; and the analysis layer that computes the evaluation metrics of Section 3.8. The bulk historical retrieval over the full target window (2015-02-19 to 2024-12-31) has likewise not yet been executed: the ingestion stage is validated on its logic and on mocked datafeed responses rather than on a completed ten-year download.

Consequently, the results enumerated in the remaining sections of this chapter (the coverage matrix of Section 4.2, the baseline and hybrid performance of Sections 4.3 and 4.4, the ablations of Section 4.5 and the comparison with the closest prior art of Section 4.6) are deferred to the final version submitted for the defense. The immediate engineering priority is the vertical completion of the price-only path, namely the transform stage, a minimal execution-engine integration and a baseline backtest, so that the first empirical figures of Section 4.3 can be produced; the textual stages follow once that path yields a result.

4.2 Data Coverage

To be added in the final version.

4.3 Baseline Strategy

To be added in the final version.

4.4 Hybrid Strategy

To be added in the final version.

4.5 Ablations

To be added in the final version.

4.6 Comparison with Closest Prior Art

To be added in the final version.

4.7 Threats to Validity Specific to the Observed Results

To be added in the final version.

5 CONCLUSION

5.1 Final Remarks

This work describes a hybrid algorithmic trading strategy for Forex that integrates technical-analysis features, indicators, chart and candlestick patterns, and NLP-derived sentiment and event signals over a ten-year corpus, designed to be evaluated under a strict walk-forward and combinatorial purged cross-validation protocol. The methodological contribution is the explicit separation of an *agentic perception layer*, in which language models and other non-deterministic components produce cached features, from a *deterministic execution layer*, in which a rule-based stage on those features emits the trade decision. The separation preserves backtest reproducibility and regulatory auditability while permitting the perception layer to incorporate increasingly capable models in subsequent iterations. The empirical assessment of this design, to the extent supported by the data and components runnable at the time of submission, is the subject of Chapter 4; the portion that remains future work is recorded explicitly in Section 5.2.

A note on the empirical conclusion. The accept/reject outcome of hypothesis H1 (Section 1.1) depends on the evidence reported in Chapter 4, which is itself sized by the methodology-freeze deadline of this manuscript. Should that evidence reject H1, the contribution of the present work remains: the methodological discipline (combinatorial purged k -fold with embargo, agentic perception with deterministic execution, configurable filter chain), the data-architecture decision (Parquet as canonical source of truth, a durable `lean-data/` execution store materialized once for replay), the coverage rule of Section 3.5.3, the license-compatible execution path of Section 3.9 and the documented conditions of failure are themselves the result. Negative results are recorded as such in Chapter 4, not hidden. A null result that survives the validation discipline of López de Prado (2018) is more informative than a positive result that does not.

5.2 Future Work

The methodology of Chapter 3 is the foundation for the subsequent research directions discussed in this section. The unifying objective behind them is to take the minimum viable product targeted by the present work to a production-grade algorithmic-trading system suitable for live capital deployment. Tiers A to D below organize the evolution of the perception layer; Section 5.2.5 discusses the eventual migration of the execution engine; Section 5.2.6 discusses the transition from backtest to paper trading to live trading; the remaining subsections extend the empirical evaluation rather than the methodology. The deterministic execution layer of Section 3.7.2 and the YAML-configurable filter chain are intended to make each extension additive: a configuration change or a new perception adapter rather than an architectural rewrite.

5.2.1 Tier A: richer single-agent perception

The simplest extension replaces the FinBERT-class scorer of Section 3.6.4 with a more capable language-model agent. A structured-output prompt against a contemporary foundation model returns, per article, a polarity score, a confidence score, and a short reasoning trace. The reasoning trace is consumed only for auditing and post-hoc analysis; the scalar polarity and confidence enter the feature stream just as FinBERT outputs do. The execution layer is unchanged. This tier requires no architectural change, only a swap of the perception module.

Empirical pedigree for the swap is recent. Kirtac and Germano (2024) compare BERT, OPT (a GPT-3-family decoder), FinBERT and the Loughran–McDonald dictionary on 965 375 U.S. financial news articles from January 2010 to June 2023, and report next-day return-prediction accuracies of 74.4 % for OPT, 72.5 % for BERT, 72.2 % for FinBERT and 50.1 % for Loughran–McDonald. The corresponding long–short portfolio Sharpe ratios are 3.05 for OPT, 2.11 for BERT and 2.07 for FinBERT. The evidence concerns U.S. equities rather than Forex, and the Sharpe magnitudes should be read under the methodological caveats of Section 3.10.1 of the methodology chapter; nevertheless, the relative ordering $\text{OPT} > \text{BERT} \approx \text{FinBERT} \gg \text{lexicon}$ is the precise pedigree that motivates the Tier A swap of the present work’s scorer for a contemporary foundation model.

5.2.2 Tier B: specialist multi-agent perception

A more ambitious extension decomposes the perception layer into specialist agents, each responsible for a particular kind of source. Concrete candidates include a monetary-policy parser specialized on FOMC, ECB and BOJ communications; a geopolitical-event-extraction agent specialized on the GDELT Global Knowledge Graph; a macroeconomic-news classifier; and a chart-context describer that converts price-series segments into a textual summary consumed by the other agents. Each agent emits structured features that are concatenated into the news-derived input matrix of Section 3.7. The deterministic execution layer is unchanged; the agents merely produce a richer perception. Concurrent work outside the Forex domain explores closely related multi-agent decompositions: TradingGroup (Tian; Salim; Xue, 2025) introduces specialist agents for sentiment analysis, report interpretation, trend forecasting and style adaptation, with a self-reflection mechanism and an automated data-synthesis pipeline that generates post-training data for the agents, and QuantAgent (Xiong *et al.*, 2025) factorises the technical-analysis layer of a high-frequency price-driven system into independent indicator, pattern, trend and risk agents. The present-work tier B differs in that the specialist agents feed cached features into the deterministic execution layer rather than emitting trade decisions directly, which preserves the validation discipline of López de Prado (2018). This tier introduces orchestration and caching concerns that the current LEAN runtime is not optimized for, and motivates the platform-migration discussion of Section 5.2.5.

5.2.3 Tier C: consensus across heterogeneous agents

A natural extension of Tier B is a consensus layer that aggregates the outputs of several agents using different methodological approaches over the same input. A sentiment score, for example, may be produced independently by a dictionary-based scorer following Loughran and McDonald (2011), by a fine-tuned BERT model in the FinBERT family (Araci, 2019; Yang; Uy; Huang, 2020), and by a foundation-model agent. Their three outputs are combined by a decision-tree-like aggregator into a single consensus score. The aggregator may be a fixed rule (majority vote, weighted average) or a learned one (a gradient-boosted tree on the agent outputs). Whichever it is, the consensus output is one more feature feeding the deterministic execution rule of Section 3.7.2; it does not become the trade decision. The methodological commitment of agentic perception with deterministic execution is preserved.

The advantage of consensus over single-agent perception is robustness: agreement among heterogeneous approaches is evidence of a real signal, while disagreement flags ambiguity and may translate, through the rule layer, into a HOLD. The disadvantage is computational cost (multiple model calls per article) and an enlarged surface area for prompt sensitivity and model drift. The cost is bounded by caching the agent outputs as Parquet artifacts, in line with the storage architecture of Section 3.3, so that the consensus is recomputed only when a new agent version is introduced.

The agentic-execution lineage of TradingAgents (Xiao *et al.*, 2024) on US equities explores a related but distinct variant, in which role-playing analyst agents reach a trade decision through structured debate rather than by feeding features into a downstream rule. The present design differs in that the consensus is a feature, not a decision, which preserves the validation discipline of López de Prado (2018) that purely agentic-execution systems struggle to honor.

5.2.4 Tier D: agent-proposed rule modifications

The most ambitious tier turns the agents into proposers of rule modifications rather than producers of features. The deterministic execution rule of Section 3.7.2 is treated as a parameterized template; an agent surveys recent backtest results, proposes a parameter change, and the change is gated by a walk-forward acceptance criterion before it is committed to the live rule layer. The execution layer remains deterministic at any point in time; the agents become rule-evolution proposers, not trade-decision makers. This tier reuses the configuration schema of the project's specifications, including the policy that distinguishes trading-impactful parameters from operational parameters, so that rule modifications are limited to the parameters the schema declares modifiable.

5.2.5 Platform migration toward an actor-based engine

The LEAN engine adopted in Section 3.9 is well-suited to the deterministic execution model of Chapter 3. As the perception layer evolves toward Tiers B, C and D, the orchestration

of multiple specialist agents (with their independent caches, model calls and lifecycles) becomes a substantial engineering concern of its own. A natural target for the next-generation platform is an actor-based engine such as NautilusTrader¹, whose architecture cleanly separates concurrent components and is used in production high-frequency settings. NautilusTrader's license is LGPL v3, which constrains commercial reuse more strictly than LEAN's Apache 2.0 license but remains compatible with academic and proprietary deployments that honor the license conditions.

The migration is not a goal of the present work. The deterministic execution rule, the configuration schema and the regression-test fixtures are, by design, engine-agnostic: they consume cached features and produce trade decisions without depending on any specific framework. Migrating the runtime therefore means re-implementing the engine-side wiring (subscriptions, indicator updates, order placement) against a different host while leaving the rule logic and the perception cache intact.

5.2.6 From MVP to production: backtest, paper trading, live capital

The artifact targeted by the present work is a minimum viable product (MVP): a self-contained pipeline that ingests price and text data, computes features under the discipline of Section 3.6, drives the deterministic filter chain of Section 3.7.2 and produces backtest results on EUR/USD (and, where the manuscript-freeze schedule allows it, USD/JPY) under the protocol of Section 3.8. The portion of the MVP that has been brought to a runnable state at the time of submission is reported in Chapter 4; portions deferred to post-publication continuation are listed explicitly in the same chapter. The MVP, in the portion that is runnable, is also the foundation for a three-stage transition to production-grade live trading. The architectural choices made in Chapter 3 were taken with that transition in mind.

The first stage retargets the same strategy code against LEAN's OANDA brokerage adapter² in paper-trading mode. This requires the run-time loader of Section 3.10 to support live-data subscriptions in addition to historical Parquet replay. Both are within the reach of LEAN's existing API and require no modification to the rule layer. The second stage runs the same code against a funded OANDA demo account, with the deterministic execution layer untouched, in order to surface integration issues (latency, partial fills, weekend gaps, rollover handling) that the historical backtest does not expose. The third stage migrates to live capital with conservative position-sizing caps, audit-log retention and a kill switch wired into the risk-guard filter (Section 3.7.2). The same filter-chain configuration drives all three stages; the only changes between them are the data source, the brokerage adapter and the capital cap.

Two methodological commitments are intended to survive the transition. First, every trade (backtest, paper or live) carries the same filter-by-filter audit trail (Section 3.7.2), so a forensic post-mortem of any live decision reproduces the reasoning end to end. Second, the perception

¹ https://github.com/nautechsystems/nautilus_trader

² The Apache 2.0 license on both LEAN and its OANDA adapter was a deciding factor in the choice of execution engine (Section 3.9). A copyleft-licensed engine would have constrained the post-publication continuation considered here.

cache of Section 3.3 is the canonical record of every model output that influenced a decision, so a regression of the entire decision sequence is reproducible even after the underlying perception models drift or are upgraded. Whether the MVP fulfills these commitments in practice is the subject of the empirical work reported in Chapter 4 and the work program of the post-publication roadmap above.

5.2.7 Knowledge-graph mediation and unsupervised news anomaly features

Two perception-layer extensions are recorded here as future work because they introduce new infrastructure rather than swap or refine an existing component. The first is a knowledge-graph-mediation pattern in the sense of Brito (2025): in place of feeding raw GDELT or news text into the LLM scorer, the ingestion pipeline materializes a structured graph (actors, locations, themes, emotional indices, Goldstein scale) and the LLM is prompted against the graph slice relevant to the pair under evaluation. The benefits are auditability of the context delivered to the model, deterministic reconstruction of the prompt at any timestamp, and a smaller per-call payload. The graph layer can be implemented either on a dedicated store (Neo4j, RedisGraph) or directly over the existing Parquet warehouse using DuckDB’s PGQ extension; the choice affects only the ingestion path and not the deterministic filter chain.

The second extension is an unsupervised anomaly-detection feature over the news stream itself, inspired by the LogBERT-style sequence-anomaly methodology of Santos (2024). Rather than scoring article-level polarity, the layer flags timestamps at which the topical or volumetric distribution of recent headlines for a given currency departs from its rolling baseline. The output is a non-directional risk-regime feature that complements the directional sentiment channel. The deterministic execution layer already accepts non-directional features at the news-context filter stage, so the integration cost is bounded.

5.2.8 Strategy-variant ablations

The methodology of Chapter 3 implements one strategy variant and evaluates it on EUR/USD and USD/JPY. A natural extension is a family of sibling variants that share the same filter chain but differ in the reward-to-risk ratio and in the trailing-stop policy. Adding each variant as an additional YAML configuration, using the inheritance semantics of the filter-chain configuration schema, and running the full evaluation pipeline against each produces an ablation study with no further methodological change. The ablation is a self-contained extension of the present work and is the most natural first follow-up.

5.2.9 Application to other financial markets

The architecture described in Chapter 3, agentic perception with deterministic execution via a configurable filter chain, is not specific to Forex. The empirical evaluation of the present work is conducted exclusively on the foreign-exchange market (EUR/USD and USD/JPY), and that is the deliberate scope of the dissertation; the design, however, is asset-class agnostic in

principle and was constructed so that adapting it to other markets is an exercise in configuration and adapter substitution rather than redesign. The market-varying constructs are isolated behind two seams. First, the tradable instrument is modeled by a common value object carrying the attributes shared by any market (symbol, quoting precision, the named price-movement unit and its size, minimum tradable size) composed with an asset-class-specific component that supplies only what is particular to that class; the Forex case populates a currency-pair component (base and quote currencies, the pip as the price-movement unit), whereas an equity, ETF, bond, futures or cryptocurrency case would populate its own component (exchange, contract specification, settlement convention) without altering the common object or its consumers. Second, the constructs that vary operationally (quoting conventions, position sizing, session calendar, margin and leverage) are encapsulated in dedicated filters (capital-management, risk-guard) and parameterized in the configuration, not hard-coded. The perception layer, the late-fusion meta-learner, the audit trail and the configuration policy carry no market-specific assumption. Should the evaluation reported in Chapter 4 demonstrate measurable value on Forex, the same methodology applies to equities, ETFs, bonds, futures, commodities or cryptocurrencies (and, more generally, to financial markets at large) by swapping the bulk-download adapter (analogous to the Dukascopy adapter) and the broker adapter (LEAN exposes adapters for Interactive Brokers, Alpaca, Coinbase and others) while keeping the perception, fusion and execution layers structurally unchanged. This generalization is contingent on the Forex evaluation succeeding; it is recorded here as a path for future research, not as a parallel commitment of the present work.

REFERENCES

- ARACI, D. **FinBERT: Financial Sentiment Analysis with Pre-trained Language Models**. 2019. ArXiv preprint arXiv:1908.10063.
- ARONSON, D. R. **Evidence-Based Technical Analysis: Applying the Scientific Method and Statistical Inference to Trading Signals**. Hoboken, NJ: John Wiley & Sons, 2007. ISBN 9780470088746.
- BAILEY, D. H. *et al.* The probability of backtest overfitting. **Journal of Computational Finance**, v. 20, n. 4, p. 39–69, 2017.
- BALLINARI, D.; MALY, J. **FX Sentiment Analysis with Large Language Models**. [S.l.], 2025. Swiss National Bank Working Paper 11/2025. Available at: https://www.snb.ch/en/publications/research/working-papers/2025/wp_2025_11.
- BAMPINAS, G. *et al.* Stocks, currencies, and geopolitical shocks: Evidence from advanced and emerging markets. **Economic Modelling**, v. 156, p. 107454, 2026.
- Bank for International Settlements. **Triennial Central Bank Survey: OTC Foreign Exchange Turnover in April 2022**. Basel, 2022. <https://www.bis.org/statistics/rpfx22.htm>.
- BAUMGARTNER, J. *et al.* The Pushshift Reddit dataset. In: **Proceedings of the International AAAI Conference on Web and Social Media**. [S.l.: s.n.], 2020. v. 14, n. 1, p. 830–839.
- BRITO, B. H. d. **Integrando Ciência da Informação e Processamento em Linguagem Natural: Uma abordagem baseada em frameworks para facilitação do compartilhamento de conhecimento organizacional**. 2025. Dissertação (Monografia de MBA em Inteligência Artificial e Big Data) — Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, SP, 2025.
- BULKOWSKI, T. N. **Encyclopedia of Chart Patterns**. 2. ed. Hoboken, NJ: John Wiley & Sons, 2005. (Wiley Trading Series). ISBN 9780471668268.
- CALDARA, D.; IACOVIELLO, M. Measuring geopolitical risk. **American Economic Review**, v. 112, n. 4, p. 1194–1225, 2022. GPR index data: <https://www.matteoiacoviello.com/gpr.htm>.
- CHAN, E. P. **Quantitative Trading: How to Build Your Own Algorithmic Trading Business**. Hoboken, NJ: John Wiley & Sons, 2008. ISBN 9780470411148.
- CHAN, E. P. **Algorithmic Trading: Winning Strategies and Their Rationale**. Hoboken, NJ: John Wiley & Sons, 2013. ISBN 1118460146.
- CHAN, E. P. **Algorithmic Trading: Winning Strategies and Their Rationale**. Hoboken, NJ: John Wiley & Sons, 2013. ISBN 9781118460146.
- DEVLIN, J. *et al.* BERT: Pre-training of deep bidirectional transformers for language understanding. In: **Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies**. Minneapolis, MN: [S.l.: s.n.], 2019. p. 4171–4186. <https://aclanthology.org/N19-1423/>.

DONG, Z.; FAN, X.; PENG, Z. **FNSPID: A Comprehensive Financial News Dataset in Time Series**. 2024. ArXiv preprint arXiv:2402.06698. <https://arxiv.org/abs/2402.06698>.

FATOUROS, G. *et al.* Transforming sentiment analysis in the financial domain with ChatGPT. **Machine Learning with Applications**, v. 14, p. 100508, 2023.

FELIX, C. **Explorando a Inteligência Artificial na Previsão de Tendências e Preços no Contexto do Swing Trading: Um Estudo de Caso para Aprimorar Estratégias de Investimento**. 2024. Dissertação (Monografia de MBA em Inteligência Artificial e Big Data) — Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, SP, 2024.

FERRARINI, J. E. A. **Uso de LLMs para a classificação de aspectos em feedbacks de desenvolvedores de software**. 2025. Dissertação (Monografia de MBA em Inteligência Artificial e Big Data) — Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, SP, 2025.

HAFEZ, P.; XIE, J. **Intraday Forex Trading Based on Sentiment Inflection Points**. [S.l.], 2013. RavenPack Research note. Posted to SSRN on 9 January 2013. Available at: <https://ssrn.com/abstract=2198816>.

HAFEZ, P. A.; XIE, J. Factoring sentiment risk into quant models. **RavenPack White Paper**, 2012. Cited in (Chan, 2013b), pp. 148–149.

HALLS-MOORE, M. L. **Successful Algorithmic Trading**. London, UK: QuantStart, 2015. Self-published, available at <https://www.quantstart.com/>.

HELDEN, B. M. V. **Uma análise fundamentalista e técnica do mercado cambial latino-americano**. 2021. Dissertação (Monografia de Graduação em Economia) — Faculdade de Economia, Administração e Contabilidade, Universidade de São Paulo, São Paulo, SP, 2021.

HENDERSON, C. **Currency Strategy: The Practitioner's Guide to Currency Investing, Hedging and Forecasting**. Chichester, UK: John Wiley & Sons, 2002. (Wiley Finance). ISBN 0470846844.

IACOVIDES, G.; ZHOU, W.; MANDIC, D. FinDPO: Financial sentiment analysis for algorithmic trading through preference optimization of LLMs. **arXiv preprint arXiv:2507.18417**, 2025. Available at: <https://arxiv.org/abs/2507.18417>.

KIRTAC, K.; GERMANO, G. Sentiment trading with large language models. **Finance Research Letters**, v. 62, p. 105227, 2024.

LAÏDI, A. **Currency Trading and Intermarket Analysis: How to Profit from the Shifting Currents in Global Markets**. Hoboken, NJ: John Wiley & Sons, 2009. (Wiley Trading). ISBN 9780470226230.

LEE, H. D. **Aplicação de métodos quantitativos no ADR/ORD Pairs trading**. 2014. Dissertação (Trabalho de Conclusão de Curso em Engenharia de Produção) — Escola Politécnica, Universidade de São Paulo, São Paulo, SP, 2014.

LEETARU, K.; SCHRODT, P. A. GDELT: Global data on events, location, and tone, 1979–2012. *In: International Studies Association Annual Convention*. San Francisco, CA: [S.l.: s.n.], 2013. <http://data.gdeltproject.org/documentation/ISA.2013.GDELT.pdf>.

LIU, X.; ZHANG, X. Geopolitical risk and currency returns. **Journal of Banking & Finance**, v. 161, p. 107097, 2024.

LO, A. W.; MAMAYSKY, H.; WANG, J. Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. **The Journal of Finance**, v. 55, n. 4, p. 1705–1765, 2000. NBER Working Paper No. 7613, March 2000.

López de Prado, M. **Advances in Financial Machine Learning**. Hoboken, NJ: John Wiley & Sons, 2018. ISBN 1119482089.

López de Prado, M. **Machine Learning for Asset Managers**. Cambridge: Cambridge University Press, 2020. ISBN 1108792898.

LOUGHRAN, T.; MCDONALD, B. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. **The Journal of Finance**, v. 66, n. 1, p. 35–65, 2011.

MELONE, A.; STATHOPOULOS, A. **Geopolitical Risk in Currency Markets**. [S.l.], 2026. Fisher College of Business Working Paper No. 2026-03-003. Date written 27 February 2026. Available at: <https://ssrn.com/abstract=6438919>.

MINER, R. C. **High Probability Trading Strategies: Entry to Exit Tactics for the Forex, Futures, and Stock Markets**. Hoboken, NJ: John Wiley & Sons, 2009.

MURPHY, J. J. **Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications**. New York: Prentice Hall Press, 1999. ISBN 0735200661.

NISON, S. **Japanese Candlestick Charting Techniques: A Contemporary Guide to the Ancient Investment Techniques of the Far East**. New York: New York Institute of Finance, 1991. ISBN 0139316507.

PAPADAKIS, C. *et al.* **ATLAS: Adaptive Trading with LLM Agents Through Dynamic Prompt Optimization and Multi-Agent Coordination**. 2025. ArXiv preprint arXiv:2510.15949. <https://arxiv.org/abs/2510.15949>.

PRADO, M. López de. The 10 reasons most machine learning funds fail. **The Journal of Portfolio Management**, v. 44, n. 6, p. 120–133, 2018.

PUGINA, R. G. **Uso de LLMs para a previsibilidade de projetos de desenvolvimento de software**. 2024. Dissertação (Monografia de MBA em Inteligência Artificial e Big Data) — Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, SP, 2024.

SAETTELE, J. **Sentiment in the Forex Market: Indicators and Strategies to Profit from Crowd Behavior and Market Extremes**. Hoboken, NJ: John Wiley & Sons, 2008. (Wiley Trading). ISBN 9780470208236.

SANTOS, D. M. **Sistema de Detecção e Categorização de Falhas de Infraestrutura**. 2024. Dissertação (Monografia de MBA em Inteligência Artificial e Big Data) — Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, SP, 2024.

TETLOCK, P. C. Giving content to investor sentiment: The role of media in the stock market. **The Journal of Finance**, v. 62, n. 3, p. 1139–1168, 2007.

TIAN, F.; SALIM, F. D.; XUE, H. **TradingGroup: A Multi-Agent Trading System with Self-Reflection and Data-Synthesis**. 2025. ArXiv preprint arXiv:2508.17565. <https://arxiv.org/abs/2508.17565>.

VASWANI, A. *et al.* Attention is all you need. In: **Advances in Neural Information Processing Systems**. Long Beach, CA: [S.l.: s.n.], 2017. v. 30, p. 5998–6008. <https://arxiv.org/abs/1706.03762>.

VEIGA, Á. d. **Aplicação de Algoritmos Heurísticos na Otimização dos Parâmetros de Trading com Pares Cointegrados**. 2022. Dissertação (Monografia de MBA em Engenharia Financeira) — Escola Politécnica, Universidade de São Paulo, São Paulo, SP, 2022.

WHITE, H. A reality check for data snooping. **Econometrica**, v. 68, n. 5, p. 1097–1126, 2000.

WU, S. *et al.* **BloombergGPT: A Large Language Model for Finance**. 2023. ArXiv preprint arXiv:2303.17564.

XIAO, Y. *et al.* **TradingAgents: Multi-Agents LLM Financial Trading Framework**. 2024. ArXiv preprint arXiv:2412.20138. First posted 28 December 2024. <https://arxiv.org/abs/2412.20138>.

XIONG, F. *et al.* **QuantAgent: Price-Driven Multi-Agent LLMs for High-Frequency Trading**. 2025. ArXiv preprint arXiv:2509.09995. <https://arxiv.org/abs/2509.09995>.

YANG, H.; LIU, X.-Y.; WANG, C. D. **FinGPT: Open-Source Financial Large Language Models**. 2023. ArXiv preprint arXiv:2306.06031.

YANG, Y.; UY, M. C. S.; HUANG, A. **FinBERT: A Pretrained Language Model for Financial Communications**. 2020. ArXiv preprint arXiv:2006.08097.

ZHANG, Y. **Interpretable Machine Learning for Macro Alpha: A News Sentiment Case Study**. 2025. ArXiv preprint arXiv:2505.16136. <https://arxiv.org/abs/2505.16136>. Source code: <https://github.com/yukepenn/macro-news-sentiment-trading>.